

Control and Estimation Co-Design[★]

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Abstract

We extend control co-design to incorporate state and parameter estimation, under the umbrella of stochastic optimal control (dual control). The proposed framework, Control and Estimation Co-Design (ContEst), parametrizes the plant dynamics, the output operator, and the noise covariances to represent design choices such as actuator tuning, sensor placement, measurement configuration, and the tuning of the estimator itself; these parameters affect both closed-loop performance and estimator accuracy. In place of the sequential plant–control–estimator pipeline, which is in general suboptimal, ContEst poses a single two-stage problem solved by first-order methods. A key enabler is that the gradient of each inner control–estimation value with respect to the design parameters is available exactly from the dual variables the inner solver already returns, by the envelope theorem; no differentiation through the optimizer or its optimality conditions is needed. We instantiate the framework in a linear–quadratic–Gaussian regime, where the control and estimation costs become two semidefinite programs whose design gradients are read from the duals of Lyapunov constraints; in a robust (H_∞) instantiation, where the same two axes become worst-case control and filtering problems solved by a pair of game Riccati equations sharing the identical envelope gradient; and in a locally linearized regime, where an extended Kalman filter yields a convex information-state model predictive control problem whose design gradients are read from the dynamics costates. We also show how sensor allocation, a discrete (mixed-integer) selection, admits a convex ℓ_1 relaxation that promotes sparse sensing.

Key words: Control co-design; Stochastic optimal control; State estimation; Semidefinite programming; Sensor placement; Sensitivity analysis.

1 Introduction

Engineering systems are still, overwhelmingly, designed *sequentially*. A plant is conceived and sized by one set of disciplines; the sensing and instrumentation are specified next; and the controller and estimator are designed last, once the plant and the measurement chain are frozen [19]. Each stage constrains the next, so decisions made early—before any closed-loop consideration—silently bound the performance achievable at the end. This division of labor is convenient and entrenched, but it is in general *not optimal*: the sequential solution is one feasible point of a larger joint problem and need not be a stationary point of it, because the plant and controller optimizations are *coupled* [17], which motivates solving them jointly [19].

Control co-design (CCD) addresses this on the control side, optimizing the plant and the controller together [19,3]—

either by *nested* strategies (an inner controller solve inside an outer plant search) or by *simultaneous* ones that optimize both at once, contrasted in [24]. The same idea appeared in process systems engineering as *integrated design and control* [33,36] and in aerospace as *simultaneous structure/control design* [30,23]. What these literatures share is a plant-plus-controller scope: the *sensing* and the *estimator* are typically fixed, or reduced to a separate sensor-selection step. Yet for any system that is not fully and noiselessly measured—essentially every real system—achievable performance is limited by what the estimator can infer, which is set by where and how well one senses. This echoes the classical input–output controllability principle that the best attainable closed-loop performance is fixed by the plant together with the sensor and actuator configuration, and cannot be recovered by the controller alone [38]. Sensor placement, precision, and configuration are design variables on the same footing as actuator authority, and they couple to the controller through the information available for feedback.

This paper proposes *Control and Estimation Co-Design* (ContEst): the design parameters governing actuation and sensing/estimation are placed in one two-stage optimization, whose second-stage objective is the stochastic optimal control (dual control) value (2)—the notion, originating

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with Feldbaum, that the optimal input must at once regulate the state and probe to reduce its uncertainty [18,6,28]. Because that cost is stochastic, it depends on the system not through the unobserved state but through its *information state*—the filtering density [26]—whose evolution explicitly involves the output equation. Optimizing it therefore improves the controller, the estimator, and their interaction at once. A caveat we make precise in Section 6: we reserve “dual control” for the exact value (2), whose input has the Bar-Shalom–Tse *dual effect* of acting on the state and on the information gathered [6,28]; the tractable eKF surrogate we optimize in the nonlinear regime is *caution-aware* (it prices estimation uncertainty through $\text{tr}(Q\Sigma_{t|t})$) but, after linearizing the information dynamics, *certainty-equivalent in the mean*—it captures *caution* (hedging against estimation uncertainty) but not *active probing* (injecting inputs deliberately to reduce it). The very prevalence of sequential design is thus an opportunity: many existing systems can be revisited and improved by a joint design pass, often without changing the hardware menu, only how its continuous parameters are chosen.

Contributions. (i) We formulate co-design of control and estimation as a single two-stage problem (Problem 1)—a design first stage over a stochastic control recourse (the second-stage control response)—expressed through the information state, so the sensing operator enters the objective explicitly (Section 3). (ii) We show that the design gradient of each inner value function is obtained *exactly* from the dual variables the inner convex solver already returns, by the envelope theorem (Theorem 1, Section 4)—which gives the gradient of an optimal-value function from the optimizer alone, without differentiating through it; no implicit differentiation and no differentiation of the Karush–Kuhn–Tucker (KKT) system is needed—only cheap model Jacobians. (iii) In the linear–quadratic–Gaussian (LQG) regime we reduce the inner problem, via the classical separation of estimation and control, to a control and an estimation semidefinite program (SDP), with gradients read from the duals of the Lyapunov constraints (Section 5). (iv) In the locally linearized regime an extended Kalman filter (eKF) lifts the problem to an information state and a convex model predictive control (MPC) problem, with gradients read from the dynamics costates (adjoint variables; Section 6). (v) We treat sensor allocation as a mixed-integer selection and show its ℓ_1 relaxation promotes sparse sensing (Section 7). Section 8 states the outer algorithm and Section 9 four co-design studies spanning spacecraft, process, and power systems.

Throughout, $\|x\|_Q^2 = x^\top Qx$, $\mathbb{E}$ is expectation, $\text{tr}(\cdot)$ the trace, and $M \succeq 0$ ($M \succ 0$) positive (semi)definiteness. The information available at time k is $\mathcal{Z}_k = (y_{0:k}, u_{0:k-1})$.

2 Background and related work

Control co-design and integrated design. [17] established that plant and controller optimization are coupled and cannot be separated without loss; [3] cast dynamic-system CCD

as multidisciplinary design optimization (MDO); and [24] systematized the nested and simultaneous solution strategies. In process systems engineering the same idea was developed largely as mixed-integer dynamic optimization under uncertainty [16], built on Biegler’s simultaneous full-discretization approach [9] and reviewed in [33,36]. In all of these the estimator and sensing layer are typically assumed fixed.

Structural co-design and information architecture. The structural-control community produced the linear-matrix-inequality (LMI) machinery on which our linear regime rests. Following early simultaneous structure/control designs [30,23], [22] alternated a control LMI with a plant/actuator-parameter LMI—a direct ancestor of our control SDP. Closest to the present work, [27] treated the *information architecture* (sensor/actuator gains) as decision variables jointly with control or estimation via LMIs, and [34,14] pushed this toward sparse H_2 sensing with guaranteed estimation-error bounds.

Sensor selection and LQG sensing co-design. A large literature treats sensor (and actuator) selection for *estimation*: convex relaxation [25], submodularity and greedy algorithms for Kalman filtering and network observability [40,42], and sparsity-promoting joint selection [15]. These covariance-based criteria are the control-theoretic face of optimal experimental design, where information is measured by convex functions of the error covariance or Fisher information; our estimation cost $\text{tr}(Q\Sigma^\varepsilon)$ is of A-optimal type. The identification community’s *least-costly experiment design* [20]—minimizing experiment cost subject to a closed-loop performance requirement—is the direct sibling of designing the noise level $V(\theta)$ against closed-loop cost, which our weight/noise co-design (Section 5.1) does inside the loop rather than as a separate identification stage. The closest work to ours is [39], who co-design sensing with LQG control through the separation structure; see also the joint scheduling separation result of [37] and the compositional view of [41]. These typically choose *which* sensors to deploy from a discrete menu, in the linear/LQG regime.

Dual control. That estimation belongs inside the control objective is the content of dual control [18]: an input has a *dual effect*, acting on the state and on the information gathered about it [6]. The exploration–exploitation reading is surveyed in [28].

Differentiable optimization and the envelope theorem. Our gradient computation is classical sensitivity analysis [10] and the envelope theorem [29] specialized to convex programs [12]. Recent work differentiates through convex programs by implicitly differentiating the KKT/cone solution map [4,1], including for bilevel problems [21]. As we stress in Section 4, for the *value* (not the minimizer) this machinery is unnecessary—and the difference is quantitative. Implicit differentiation of the KKT/cone map solves an $m \times m$ linear system per design direction, with $m = O(r_x^2)$ for the covariance SDPs, i.e. $O(r_x^6)$ per directional derivative, and

it differentiates the *minimizer*, which ContEst never needs; the envelope route reads the already-computed dual and contracts at $O(r_x^2 r_\theta)$ for the *whole* gradient (Section 9.5).

Positioning: closest work and what is new. Three lines of work come closest, and ContEst departs from each in the same structural way. (i) *Sensor and estimator design* [25,40,42] places or selects sensors to sharpen a Kalman-filtering or observability criterion, but for a *fixed* plant and controller: the actuation and the closed-loop control are not co-optimized with the sensing. (ii) *Plant–controller co-design* [17,24,22] jointly tunes plant and controller, but treats sensing and estimation as given—full-state feedback or a fixed observer—so the output map never enters the design. (iii) *Dual and adaptive control* [18,6,28] couples control with online estimation and probing, but for a *fixed* physical and sensing design θ : the distinction is *offline hardware vs. online policy*—dual/adaptive MPC adapts the *inputs* for a fixed design, whereas ContEst optimizes the *design* against the (approximated) closed-loop information dynamics. The nearest single reference is LQG sensing co-design [39], which co-selects sensors with an LQG controller through separation, but only in the linear/LQG regime and as a discrete menu selection; Section 9.3 compares ContEst head-to-head against the separation-style sequential pipeline that such menu methods induce.

Table 1 makes these deltas auditable per capability.

Against these, ContEst is, to our knowledge, the first to close all three loops at once—plant, controller, *and* estimator—inside a single stochastic (dual-control) objective expressed through the information state, so that estimation quality is *optimized* rather than assumed. Its distinguishing ingredients are: actuation *and* sensing/estimation parameters co-designed as *continuous* variables in one problem (with discrete sensor allocation recovered by a sparse ℓ_1 relaxation, Section 7); a nonlinear extension via the eKF and certainty equivalence [31]; and design gradients obtained *exactly* from the inner dual variables by the envelope theorem, without implicit differentiation of the solution map and without differentiating the KKT system (Section 4).

Significance. Since essentially every deployed system was designed sequentially, it sits at a feasible but generally non-stationary point of this joint problem; a single co-design pass can therefore improve it—often without altering the hardware menu, only how its continuous parameters are chosen. The studies of Section 9 bear this out, with total-cost reductions of 6–43% in which the sensing axis θ_h —absent from control-only co-design—is a major, and sometimes dominant, contributor, up to and including the tuning of the estimator itself (a phase-locked loop, Section 9.3).

3 Problem formulation

Consider the parametrized stochastic system

$$x_{k+1} = f(x_k, u_k, w_k; \theta), \quad (1a)$$

$$y_k = h(x_k, v_k; \theta), \quad (1b)$$

with state $x_k \in \mathbb{R}^{r_x}$, input $u_k \in \mathbb{R}^{r_u}$, and output $y_k \in \mathbb{R}^{r_y}$. The maps f, h are parametrized by $\theta \in \mathbb{R}^{r_\theta}$, representing plant parameters, actuator gains, and sensor tuning/placement. The disturbances w_k, v_k are i.i.d., mutually independent and independent of $x_0 \sim p_0$ (the prior, before any measurement including y_0).

The goal is to minimize design, control, and estimation costs together. The *design cost* $J_{\text{des}}(\theta)$ and design set $\theta \in \Theta$ encode the price, manufacturability, or placement difficulty of a θ . The *control and estimation cost* is the stochastic optimal control value

$$J_{\text{stoch}}(p_0; \theta) = \min_{u_0:N-1} \frac{1}{N} \mathbb{E} \left\{ \sum_{k=0}^N \ell(x_k, u_k) + \ell^N(x_N) \right\}, \quad (2)$$

subject to constraints $u_k \in \mathbb{U}$, $x_k \in \mathbb{X}$, where the expectation is over $(x_0, w_{0:N-1}, v_{0:N})$ and ℓ, ℓ^N are stage and terminal costs. In general $x_k \in \mathbb{X}$ is a chance constraint $\mathbf{P}(x_k \in \mathbb{X}) \geq 1 - \delta$; in this paper we impose the input constraints and the mean-state constraints in their deterministic (nominal) form within the inner convex programs of Sections 5–6 and the studies of Section 9, and defer a full probabilistic treatment—which enters the inner problem as a convex second-order-cone tightening of $\mathbb{X}$ and leaves the envelope machinery of Section 4 intact—to future work (Section 10).

Problem 1 (ContEst).

Design (first stage): $\min_{\theta \in \Theta} J(\theta) = J_{\text{des}}(\theta) + J_{\text{stoch}}(p_0; \theta)$,

Control (second stage): $J_{\text{stoch}}(p_0; \theta)$ as in (2).

Remark 2 (Operating scenarios). For clarity Problem 1 is stated for a single prior p_0 . Multiple *operating scenarios*—distinct initial-condition densities $p_0^{(s)}$, load levels, disturbance statistics, or tracking set-points—are accommodated verbatim by replacing the second-stage value with a weighted average

$$J_{\text{stoch}}(\theta) = \sum_{s=1}^M \pi_s J_{\text{stoch}}(p_0^{(s)}; \theta), \quad \pi_s \geq 0, \sum_{s=1}^M \pi_s = 1, \quad (3)$$

over the M scenarios with weights π_s (e.g. $\pi_s = 1/M$ for a plain average, or empirical probabilities of each regime). Each $J_{\text{stoch}}(p_0^{(s)}; \theta)$ is an independent second-stage solve and the design objective is linear in these values, so the

Table 1

Positioning of ContEst against the nearest lines of work, by capability (✓ present, ✗ absent, *partial* limited). “Estimator itself designed” means tuning a *dynamic* estimator (e.g. a PLL bandwidth, or the noise level $V(\theta)$), not only choosing sensors.

capability	CCD/MDO [19,24]	sensor selection [25,40]	LQG sensing co-design [39]	info-arch. LMIs [27]	dual/adaptive MPC [28]	diff. opt. layers [4,1]	ContEst
plant / actuation designed	✓	✗	✗	partial	✗	✗	✓
sensing / output map designed	✗	✓	✓ (menu)	✓	✗	✗	✓ (cont. + ℓ_1 menu)
estimator itself designed	✗	✗	✗	✗	✗	✗	✓
closed-loop control in objective	✓	✗	✓	partial	✓ (online)	✓	✓
nonlinear regime	✓	✗	✗	✗	✓	✓	✓ (eKF surrogate)
gradient w.r.t. design	often FD	greedy/relax.	menu enum.	alt. LMIs	n/a	implicit KKT diff.	exact from duals

envelope gradient (Section 4) is the same convex combination $\sum_s \pi_s \partial_\theta J_{\text{stoch}}(p_0^{(s)}; \theta)$ of the per-scenario gradients: nothing in the machinery changes, only the number of inner solves per design. The studies of Section 9 use a single scenario ($M = 1$).

From bilevel to two-stage. Problem 1 is, in general form, a *bilevel* program: an outer optimization over θ whose objective contains the argmin of an inner optimization over the control. Solving a bilevel problem directly is hard—it couples the two levels through the inner *minimizer* $u^*(\theta)$, whose sensitivity $\partial u^*/\partial \theta$ requires implicitly differentiating the inner optimality conditions (an MPEC). The structure here, however, collapses this: the design θ is a *here-and-now* decision fixed before any realization, and the control is the *recourse* that reacts to the information $\mathcal{Z}_k$, so Problem 1 is precisely a *two-stage stochastic program* with first-stage variable θ and second-stage (recourse) value $J_{\text{stoch}}(p_0; \theta)$. What the outer method needs from the second stage is not its minimizer but only the sensitivity of its *value*, and for that the envelope theorem (Section 4) supplies the gradient directly from the duals the inner solver already returns—no differentiation of $u^*(\theta)$ and no MPEC. It is this reduction from a bilevel to a two-stage problem that makes ContEst tractable, and we treat it as two-stage throughout.

Compared with finite-difference sensitivities, ContEst obtains the full design gradient at essentially the cost of a single inner solve: forward (central) differences need $r_\theta + 1$ ($2r_\theta$) inner solves for an r_θ -vector θ , whereas ContEst solves the inner problem once and contracts its returned duals or costates against model Jacobians—an $O(r_\theta)$ speedup—and returns the *exact* gradient of the inner value rather than a truncated, step-size sensitive approximation.

3.1 The information state

The second stage is a stochastic (dual) control problem [18,6]. A causal law must be a function of the available information $\mathcal{Z}_k$, not of the unobserved x_k ; the sufficient statistic is the filtering density $p(x_k | \mathcal{Z}_k; \theta)$, governed by the Bayesian filter. With $\mathcal{Z}_k = (y_{0:k}, u_{0:k-1})$, the filter alter-

nates a time update,

$$p(x_{k+1} | \mathcal{Z}_k, u_k) = \int p(x_{k+1} | x_k, u_k; \theta) p(x_k | \mathcal{Z}_k) dx_k, \quad (4)$$

and a measurement update,

$$p(x_{k+1} | \mathcal{Z}_{k+1}) \propto p(y_{k+1} | x_{k+1}; \theta) p(x_{k+1} | \mathcal{Z}_k, u_k). \quad (5)$$

Both (4)–(5) involve the output equation (1b) and hence the sensing parameters in θ . The standard smoothing identity makes this explicit in the cost.

Lemma 3 (Smoothing [32, Ch. 10]). *For a probability space $(\Omega, \mathcal{A}, \mathbf{P})$, a measurable $\chi : \Omega \rightarrow \mathbb{R}$ with $\mathbb{E}_{\mathbf{P}}|\chi| < \infty$, and σ -algebras $\mathcal{A}_0 \subset \mathcal{A}_1 \subset \mathcal{A}$, $\mathbb{E}_{\mathbf{P}}\{\chi | \mathcal{A}_1\} = \mathbb{E}_{\mathbf{P}}\{\mathbb{E}_{\mathbf{P}}\{\chi | \mathcal{A}_0\} | \mathcal{A}_1\}$.*

Lemma 4. *For a causal law $u_k = u_k(\mathcal{Z}_k, p_0)$,*

$$\mathbb{E} \ell(x_k, u_k) = \int \int \ell(x_k, u_k) p(x_k | \mathcal{Z}_k; \theta) p(\mathcal{Z}_k; \theta) dx_k d\mathcal{Z}_k.$$

Proof. Conditioning on $\mathcal{Z}_k$ yields a σ -algebra contained in the one underlying (2); apply Lemma 3. $\square$

Lemma 4 shows the cost depends on the anticipated filtering densities, hence on the output equation: optimizing it improves the controller, the filter, and their coupling. (Causality requires conditioning on $\mathcal{Z}_{k'}$ with $k' \leq k$; we take $k' = k$.)

4 Exact design gradients via the envelope theorem

The two-stage problem is solved by descent on the first-stage variable θ , so the crux is $\nabla_\theta J_{\text{stoch}}$: the sensitivity of a *value function* to parameters entering its objective and constraints. Our central observation is that this gradient is essentially free—and it is exactly this that reduces the bilevel program to the two-stage one. Write the convex second-stage problem in conic form,

$$V(\theta) = \min_z \phi(z, \theta) \quad \text{s.t.} \quad g(z, \theta) = 0, \quad G(z, \theta) \in \mathcal{K}, \quad (6)$$

with closed convex cone $\mathcal{K}$ (the positive semidefinite cone for SDPs), Lagrangian $L(z, \mu, S, \theta) = \phi(z, \theta) + \langle \mu, g(z, \theta) \rangle + \langle S, G(z, \theta) \rangle$, and optimal primal–dual triple (z^*, μ^*, S^*) . We collect the hypotheses under which the value V is differentiable and its gradient is read from that triple.

Assumption 1. On a neighborhood Θ_0 of the design point θ : (A1) for each $\theta \in \Theta_0$, (6) is convex in z , its data (ϕ, g, G) are jointly C^1 in (z, θ) , and Slater’s condition holds, so strong duality holds with an attained primal–dual solution; (A2) the primal minimizer $z^*(\theta)$ is unique; (A3) the primal–dual solution is *nondegenerate* and strictly complementary, so the dual (μ^*, S^*) is unique [2,35]; (A4) θ enters (6) only through finitely many data objects $D(\theta)$ (e.g. $A_\theta, B_\theta, C_\theta, W, V$) that are C^1 in θ .

Assumptions (A1)–(A3) are exactly the regularity under which the optimal value of a conic (in particular semidefinite) program is differentiable and its sensitivity is governed by the unique dual [10,35]; (A4) is a modeling convenience that isolates the θ -dependence in cheap model Jacobians.

Theorem 1 (Exact design gradient from the inner dual). Under Assumption 1, V is differentiable at θ and

$$\nabla_\theta V(\theta) = \nabla_\theta L(z^*, \mu^*, S^*, \theta) = \partial_\theta \phi + \langle \mu^*, \partial_\theta g \rangle + \langle S^*, \partial_\theta G \rangle, \quad (7)$$

all partials evaluated at (z^*, θ) with z^*, μ^*, S^* held fixed.

Proof. By (A1)–(A3) the solution map $\theta \mapsto (z^*, \mu^*, S^*)$ is single-valued and differentiable on Θ_0 [10,35], and strong duality with complementary slackness gives $V(\theta) = L(z^*(\theta), \mu^*(\theta), S^*(\theta), \theta)$. Differentiating and using inner stationarity $\nabla_z L = 0$ and the stationarity of L in (μ, S) at the solution, $\nabla_\theta V = \nabla_z L \partial_\theta z^* + \nabla_{(\mu, S)} L \partial_\theta (\mu^*, S^*) + \partial_\theta L = \partial_\theta L$ [29,10]. $\square$

Remark 5 (Degenerate case). If (A3) fails, the dual solution set $\mathcal{D}^*(\theta)$ is not a singleton and V is in general only directionally differentiable, with $V'(\theta; d) = \min_{(\mu, S) \in \mathcal{D}^*} \langle \nabla_\theta L(z^*, \mu, S, \theta), d \rangle$ [10]. The right-hand side of (7) evaluated at *any* returned dual is then a subgradient rather than the gradient, and the outer method of Section 8 degrades gracefully to a subgradient step. In our studies strict complementarity held at every solution (consistent with the gradient check of Section 9). By Proposition 1, in the LQG regime degeneracy can arise only where that proposition’s hypotheses fail (e.g. a singular process noise W , or a loss of detectability at isolated θ), which pinpoints in advance the design points that need the subgradient fallback.

Three points give ContEst its practical edge. (1) *No implicit differentiation:* (7) has no term $\partial_\theta z^*$; we never compute how the minimizer moves with θ , which is precisely what implicit-function-theorem methods spend effort on [4,1]. (2) *No KKT differentiation:* the multipliers (μ^*, S^*) are not

recomputed by differentiating optimality conditions; they are the dual solution the inner solver already returns, and we read them off. (3) *Only model Jacobians remain:* in our setting θ enters (6) through a small set of data objects D (e.g. the matrices $A_\theta, B_\theta, \dots$), so $\nabla_\theta V = \sum_D \langle \partial V / \partial D, \partial D / \partial \theta \rangle$, where $\partial V / \partial D$ comes from (7) and $\partial D / \partial \theta$ are explicit (or automatically differentiated) Jacobians of the *model*, not the optimizer.

5 The LQG case: LQR and Kalman SDPs

Let (1) be linear (or linearized about an operating point), $x_{k+1} = A_\theta x_k + B_\theta u_k + w_k$, $y_k = C_\theta x_k + v_k$, with zero-mean white w_k, v_k of covariances W and V (possibly functions of θ). With the observer $\hat{x}_{k+1} = A_\theta \hat{x}_k + B_\theta u_k + G(y_k - C_\theta \hat{x}_k)$, $A_\theta - GC_\theta$ stable, the error $\varepsilon_k = x_k - \hat{x}_k$ obeys $\varepsilon_{k+1} = (A_\theta - GC_\theta)\varepsilon_k + w_k - Gv_k$. For $J_\theta = \lim_{k \rightarrow \infty} \mathbb{E}\{x_k^\top Q x_k + u_k^\top R u_k\}$ with $u_k = K \hat{x}_k$ and $x_k = \hat{x}_k + \varepsilon_k$, the cyclic property of the trace gives, with $\Sigma = \text{Cov}(\hat{x}_k)$ and $\Sigma^\varepsilon = \text{Cov}(\varepsilon_k)$,

$$J_\theta = \underbrace{\text{tr}(Q\Sigma^\varepsilon)}_{J_{\text{est}}(\theta)} + \underbrace{\text{tr}(Q\Sigma) + \text{tr}(RK\Sigma K^\top)}_{J_{\text{cont}}(\theta)}. \quad (8)$$

This is the separation principle, and we use it *only within the inner solve*: for a fixed θ , J_{est} and J_{cont} depend on disjoint variables and may be minimized independently [39]. It is *not* a design-level separation. At the design level θ couples $A_\theta, B_\theta, C_\theta$ and the noise covariances, hence both the control gain K and the observer gain G , so $J_{\text{est}}(\theta)$ and $J_{\text{cont}}(\theta)$ vary over a common θ and the co-design objective is *not* separable across the two axes—which is precisely why the joint two-stage formulation of Problem 1 is needed rather than a place-then-control pipeline.

Control SDP. Following the output-variance-constrained construction [22,12],

$$\begin{aligned} J_{\text{cont}}(\theta) &= \min_{\Sigma, L, Z_0} \text{tr}(Q\Sigma) + \text{tr}(RZ_0) \\ \text{s.t. } \Sigma &\succ 0, \quad \begin{bmatrix} Z_0 & L \\ L^\top & \Sigma \end{bmatrix} \succeq 0, \\ &\quad \begin{bmatrix} \Sigma - W & A_\theta \Sigma + B_\theta L \\ (A_\theta \Sigma + B_\theta L)^\top & \Sigma \end{bmatrix} \succeq 0, \end{aligned} \quad (9)$$

with optimal gain $K^* = L^*(\Sigma^*)^{-1}$.

Estimation SDP. The error covariance solves a Lyapunov equation that, after $F = P^\varepsilon G$ and the LMI relaxation, gives

$$\begin{aligned} J_{\text{est}}(\theta) &= \min_{P^\varepsilon, F, Z_1} \text{tr}(WP^\varepsilon) + \text{tr}(VZ_1) \\ \text{s.t. } P^\varepsilon &\succ 0, \quad \begin{bmatrix} Z_1 & F^\top \\ F & P^\varepsilon \end{bmatrix} \succeq 0, \\ &\quad \begin{bmatrix} P^\varepsilon - Q & A_\theta^\top P^\varepsilon - C_\theta^\top F^\top \\ P^\varepsilon A_\theta - FC_\theta & P^\varepsilon \end{bmatrix} \succeq 0, \end{aligned} \quad (10)$$

dual in form to (9) and exposing the sensing operator C_θ , i.e. the estimation half of the co-design [27].

Design gradients from the LMI duals. Both (9) and (10) are instances of (6). Applying Theorem 1 to the Lyapunov constraint of (9), whose dual we partition as $S = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^\top & S_{22} \end{bmatrix} \succeq 0$, gives

$$\frac{\partial J_{\text{cont}}}{\partial A_\theta} = -2 S_{12} \Sigma^*, \quad \frac{\partial J_{\text{cont}}}{\partial B_\theta} = -2 S_{12} (L^*)^\top, \quad (11)$$

with analogous expressions for W, Q, R from the remaining Lagrangian terms, and a parallel set for (10). The design gradient follows by the chain rule of Theorem 1(3),

$$\frac{\partial J_{\text{cont}}}{\partial \theta_i} = \text{tr} \left(\frac{\partial J_{\text{cont}}}{\partial A_\theta} \frac{\partial A_\theta}{\partial \theta_i} \right) + \text{tr} \left(\frac{\partial J_{\text{cont}}}{\partial B_\theta} \frac{\partial B_\theta}{\partial \theta_i} \right) + \dots, \quad (12)$$

where the model Jacobians $\partial A_\theta / \partial \theta_i$, etc., are cheap and exact. No sweep through the SDP solver is required.

Rather than *assume* the regularity of Assumption 1 for these SDPs, we can *verify* it from the model data. The following makes the hypotheses checkable and, in doing so, identifies the LMI duals as classical Riccati objects.

Proposition 1 (LQG regularity). Fix θ and suppose (A_θ, B_θ) is stabilizable, (A_θ, C_θ) is detectable, and $Q, R, W, V \succ 0$, with all data C^1 in θ . Then for the control SDP (9) and the estimation SDP (10): (a) Slater’s condition holds; (b) the primal minimizer is unique— Σ^* the stabilizing closed-loop covariance and $K^* = L^*(\Sigma^*)^{-1}$ the Riccati gain, and dually $P^{\varepsilon*} = \Sigma^\varepsilon$ the Kalman error covariance; (c) the Lyapunov-LMI dual is unique and strictly complementary—in fact the $(1, 1)$ dual block of the control LMI is the control Riccati solution P (so $S_{11}^* = P$) and the $(1, 1)$ dual block of the estimation LMI is the steady-state Kalman error covariance Σ^ε (so $S_{11}^{\varepsilon*} = \Sigma^\varepsilon$). Hence Assumption 1 holds on a neighborhood Θ_0 of θ , and Corollary 1 applies.

Proof sketch. A strictly feasible point is built from any stabilizing gain inflated by εI (Slater); with $W \succ 0$ a Lyapunov-eigenvector argument forces every feasible Σ to encode a stabilizing gain, and the completion-of-squares identity $J(K) - \text{tr}(PW) = \text{tr}((K - K^*)^\top (R + B^\top P B)(K - K^*) \Sigma_K)$ gives primal uniqueness; complementary slackness reduces the dual stationarity to the closed-loop Stein equation $\Lambda = Q + K^{*\top} R K^* + A_{\text{cl}}^\top \Lambda A_{\text{cl}}$, whose unique solution is $\Lambda = P$, and strict complementarity follows from rank counting with $P \succeq Q \succ 0$, $R \succ 0$. The full argument and the dual identities are in Appendix A. $\square$

Controllability and observability imply the stabilizability/detectability hypotheses, so the practical reading of Proposition 1 is: *controllable, observable plant with positive-definite weights $\Rightarrow$ everything in Sections 4–5 is exact and differentiable*. It also explains structurally why the Riccati gradient formulas of Section 9.5 coincide with the LMI-dual formulas (13)–(14): they are literally the same object, $S_{11}^* = P$.

Corollary 1 (LQG design gradient). Under the hypotheses of Proposition 1 (stabilizability, detectability, $Q, R, W, V \succ 0$), the SDPs (9)–(10) satisfy Assumption 1, so the Lyapunov-LMI dual S (resp. its estimation counterpart) is unique. Then $J_{\text{cont}}(\theta)$ and $J_{\text{est}}(\theta)$ are differentiable at θ and their design gradients are given exactly by (11)–(12) (resp. the parallel expressions in C_θ, W, V), read from the returned LMI duals.

5.1 Co-designing the weights and noise levels

Nothing in the framework restricts θ to the dynamics and output maps. The weights Q, R of the quadratic cost and the noise covariances W, V of the process and measurement channels are simply more of the data objects $D(\theta)$ of Assumption 1(A4), so making any of them a design variable requires *no new machinery*: the same duals contract against a new model Jacobian. This matters in practice because the noise levels are exactly the knobs a sensor or actuator budget buys—a better inertial unit lowers V , a stiffer mount or a cleaner drive lowers W —so “how good a sensor to buy” is a continuous co-design variable on the same footing as “where to place it.”

Concretely, both LQG subproblems are affine in these data, so their sensitivities are read off directly, with no LMI-dual partition required for the objective terms. From the control SDP (9),

$$\frac{\partial J_{\text{cont}}}{\partial Q} = \Sigma^*, \quad \frac{\partial J_{\text{cont}}}{\partial R} = Z_0^*, \quad \frac{\partial J_{\text{cont}}}{\partial W} = -S_{11}^*, \quad (13)$$

the first two from the objective $\text{tr}(Q\Sigma) + \text{tr}(RZ_0)$ and the last from the $(1, 1)$ block $\Sigma - W$ of the Lyapunov LMI, whose dual is S_{11}^* . Dually, from the estimation SDP (10),

$$\frac{\partial J_{\text{est}}}{\partial W} = P^{\varepsilon*}, \quad \frac{\partial J_{\text{est}}}{\partial V} = Z_1^*, \quad \frac{\partial J_{\text{est}}}{\partial Q} = -S_{11}^{\varepsilon*}, \quad (14)$$

with $S_{11}^{\varepsilon*}$ the $(1, 1)$ dual block of the estimation LMI. Two structural facts are worth noting. First, Q and W each appear in *both* subproblems— Q weighs the control cost $\text{tr}(Q\Sigma)$ and the estimation cost $\text{tr}(Q\Sigma^\varepsilon)$, and W is both the process noise seen by the estimator and a term in the control Lyapunov constraint—so the total gradient is just the sum of the two contributions, e.g. $\partial J_\theta / \partial W = -S_{11}^* + P^{\varepsilon*}$. Second, these expressions substitute into the same chain rule (12): whatever the underlying design variable θ_i that moves Q, R, W or V , its gradient is $\text{tr}((\partial J / \partial Q)^\top \partial Q / \partial \theta_i) + \dots$, with cheap model Jacobians. The eKF case of Section 6 is identical in spirit: Q, R enter the stage cost ℓ directly, while the noise covariances Σ_w, Σ_v enter the filter map and hence $A_\theta^{\text{info}}, B_\theta^{\text{info}}$, so their gradients are delivered by the costate contraction (24) with no change.

A caveat keeps the extension honest. Designing W, V is unambiguous—they are physical noise levels priced by J_{des} ,

and lowering them always helps the inner value, so the design cost is what makes the trade-off nontrivial. Designing Q, R is meaningful only when the weights are genuinely negotiable knobs—an actuation-effort budget, or a specification the designer is free to retune—rather than a fixed performance specification; absent a competing term in J_{des} , minimizing the closed-loop cost over its own weights is degenerate (e.g. $R \rightarrow 0$). We therefore treat Q, R as design variables only against an explicit J_{des} penalty or bound, exactly as for the other θ coordinates.

5.2 The robust analog: H_∞ control and filtering SDPs

The LQG development above prices *average* performance against *stochastic* white disturbances. When the disturbances are better modeled as unknown-but-bounded ℓ_2 signals and the designer cares about the *worst case*, the same linear plant admits an H_∞ inner problem, and — crucially for ContEst — it again reduces to a pair of semidefinite programs of the generic form (6), so Theorem 1 delivers the design gradient with no new machinery. The inner *value* is now the squared worst-case closed-loop ℓ_2 -gain $\gamma^2(\theta)$ rather than a stationary variance, but it is still the optimal value of a conic program whose θ -dependence is confined to the model data.

Write the linear plant with an explicit disturbance channel and a performance output,

$$x_{k+1} = A_\theta x_k + B_\theta u_k + E_\theta w_k, \quad z_k = C_z x_k + D_z u_k, \quad (15)$$

where $w_k \in \ell_2$ is the exogenous disturbance, E_θ its input map (e.g. $E_\theta = W_\theta^{1/2}$, so the process “noise level” of the LQG model becomes a disturbance-channel weight), and z_k stacks the penalized state and input, $C_z = [Q^{1/2}; 0]$, $D_z = [0; R^{1/2}]$, so the H_∞ objective weighs exactly the same signals as J_{cont} .

Control SDP. By the discrete-time bounded-real lemma with the standard linearizing change of variables $Y = P^{-1} \succ 0$, $L = KY$ [11,22], the state-feedback H_∞ synthesis problem is the SDP (with $\mu := \gamma^2$)

$$\begin{aligned} \gamma_{\text{cont}}^2(\theta) = \min_{Y, L, \mu} \quad & \mu \\ \text{s.t. } Y \succ 0, \quad & \begin{bmatrix} -Y & \star & \star & \star \\ 0 & -\mu I & \star & \star \\ A_\theta Y + B_\theta L & E_\theta & -Y & \star \\ C_z Y + D_z L & 0 & 0 & -I \end{bmatrix} \preceq 0, \end{aligned} \quad (16)$$

where $\star$ denotes the block inferred by symmetry, with optimal gain $K^\star = L^\star(Y^\star)^{-1}$. This is the exact robust counterpart of the output-variance-constrained control SDP (9): the Lyapunov certificate Y plays the role of the covariance Σ , the extra disturbance row/column carries the gain bound μ , and the performance block $C_z Y + D_z L$ replaces the traced objective.

Estimation SDP. Dually, for the filter $\hat{x}_{k+1} = A_\theta \hat{x}_k + G(y_k - C_\theta \hat{x}_k)$ with measurement $y_k = C_\theta x_k + D_v v_k$ and estimation-error output $\zeta_k = H(x_k - \hat{x}_k)$, the bounded-real lemma applied to the error dynamics — now affine in (P, F) under $F = PG$ — gives the H_∞ filtering SDP

$$\begin{aligned} \gamma_{\text{est}}^2(\theta) = \min_{P, F, \mu} \quad & \mu \\ \text{s.t. } P \succ 0, \quad & \begin{bmatrix} -P & \star & \star & \star \\ A_\theta^\top P - C_\theta^\top F^\top & -P & \star & \star \\ E_\theta^\top P - D_v^\top F^\top & 0 & -\mu I & \star \\ 0 & H & 0 & -I \end{bmatrix} \preceq 0, \end{aligned} \quad (17)$$

with optimal gain $G^\star = (P^\star)^{-1}F^\star$, dual in form to (16) and exposing the sensing operator C_θ and measurement weight $D_v = V_\theta^{1/2}$ — the estimation half of the robust co-design [27]. As in the LQG case the fixed- θ separation is an inner convenience: at the design level θ couples $A_\theta, B_\theta, C_\theta, E_\theta, D_v$ across *both* LMIs, so the worst-case co-design objective is again non-separable and Problem 1 is needed.

Design gradients from the BRL duals. Both (16) and (17) are instances of (6): the constraint is a single conic inclusion $-\mathcal{M}(\cdot, \theta) \in \mathcal{K}$ (the PSD cone), so Theorem 1 gives $\nabla_\theta \gamma^2 = -\langle S^\star, \partial_\theta \mathcal{M} \rangle$, with $S^\star \succeq 0$ the returned BRL dual and $\partial_\theta \mathcal{M}$ nonzero only in the blocks that carry θ . Partition the control dual conformally with the 4×4 LMI of (16), $S^\star = [S_{ij}]$, $S_{ij} = S_{ji}^\top$. The plant blocks $A_\theta Y + B_\theta L$ sit in position (3, 1), giving

$$\frac{\partial \gamma_{\text{cont}}^2}{\partial A_\theta} = -2 S_{13} Y^\star, \quad \frac{\partial \gamma_{\text{cont}}^2}{\partial B_\theta} = -2 S_{13} (L^\star)^\top, \quad (18)$$

with analogous expressions for C_z, D_z (from the (4, 1) block S_{14}) and for E_θ (from the (3, 2) block S_{23}), and structurally identical to the LQG control gradient (11) with the covariance dual S_{12} replaced by the BRL dual S_{13} . For the filter, partitioning the dual of (17) as $\tilde{S} = [\tilde{S}_{ij}]$, the off-diagonal block carrying $PA_\theta - FC_\theta$ yields

$$\frac{\partial \gamma_{\text{est}}^2}{\partial A_\theta} = -2 P^\star \tilde{S}_{12}, \quad \frac{\partial \gamma_{\text{est}}^2}{\partial C_\theta} = 2 (F^\star)^\top \tilde{S}_{12}, \quad (19)$$

with parallel terms in E_θ, D_v . In both cases the design gradient closes through the same chain rule (12) against cheap model Jacobians $\partial A_\theta / \partial \theta_i, \partial C_\theta / \partial \theta_i, \dots$; as before, no sweep through the SDP solver is required.

Corollary 2 (H_∞ design gradient). Let (16) (resp. (17)) satisfy Assumption 1 — for these SDPs, strict feasibility (a stabilizing gain achieving a gain strictly below the optimum, i.e. Slater) and nondegeneracy of the optimal solution [2], so the BRL dual $S^\star$ (resp. $\tilde{S}^\star$) is unique. Then $\gamma_{\text{cont}}^2(\theta)$ and $\gamma_{\text{est}}^2(\theta)$ are differentiable at θ and their design gradients are

given exactly by (18) (resp. (19)) and their analogs, read from the returned BRL duals. If strict complementarity fails, Remark 5 applies verbatim and the returned dual yields a subgradient.

Remark 6 (Where the SDP route is exact, and where it is not). The two regularity regimes must be kept separate. At a *fixed* γ above the optimal attenuation the story is clean and is stated as Proposition 2 below (the fixed- γ game Riccati is what the robust study of Section 9.4 actually uses). At the *minimal* $\gamma = \gamma^*$ that (16) minimizes over, by contrast, the BRL LMI loses rank and strict complementarity *generically fails*, so only the subgradient statement of Remark 5 is available; this is why we never differentiate at γ^* and prefer the fixed- γ Riccati form for numerics.

Scalable game-Riccati route. Exactly as the H_2 control SDP (9) has an $O(r_x^3)$ Riccati twin ($J_{\text{cont}} = \text{tr}(PW)$) from the control DARE, used for the large-scale study of Section 9.4), the H_∞ control SDP admits a Riccati twin that avoids the interior-point solve entirely — important because, at the *minimal*- γ optimum, the BRL LMI (16) loses rank and its dual is numerically delicate. Fix the attenuation level γ and pose the soft-constrained dynamic game $\min_u \max_w \sum_k (x_k^\top Q x_k + u_k^\top R u_k - \gamma^2 w_k^\top w_k)$ for the plant (15) with $E_\theta = W_\theta^{1/2}$; its value is $x_0^\top X x_0$ with $X \succeq 0$ the stabilizing solution of the discrete H_∞ game Riccati equation. Writing $\tilde{B}_\theta = [B_\theta \ E_\theta]$ and the *indefinite* game weight $\tilde{R} = \text{diag}(R, -\gamma^2 I)$, that GARE is precisely a discrete algebraic Riccati equation in the augmented data $(A_\theta, \tilde{B}_\theta, Q, \tilde{R})$,

$$X = Q + A_\theta^\top X A_\theta - A_\theta^\top X \tilde{B}_\theta (\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta, \quad (20)$$

admissible when $R + B_\theta^\top X B_\theta \succ 0$ and the game closed loop $A_{\text{cl}} = A_\theta + \tilde{B}_\theta \tilde{K}$, $\tilde{K} = -(\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta = [K; K_w]$, is Schur. The *worst-case (guaranteed) stationary cost*

$$V_{\text{wc}}(\theta) = \text{tr}(XW_\theta) \quad (21)$$

is the robust analog of $\text{tr}(PW)$, and as $\gamma \rightarrow \infty$ the disturbance player switches off, $X \rightarrow P$ and $V_{\text{wc}} \rightarrow \text{tr}(PW)$, the H_2 cost. Crucially, (20) is a standard DARE (only its input weight is indefinite), so *the envelope-theorem sensitivities are the very same algebraic identities as the H_2 Riccati*—an identity that never used $R \succ 0$. With S the game-closed-loop gramian $S = A_{\text{cl}} S A_{\text{cl}}^\top + W_\theta$,

$$\frac{\partial V_{\text{wc}}}{\partial A_\theta} = 2 X A_{\text{cl}} S, \quad \frac{\partial V_{\text{wc}}}{\partial \tilde{B}_\theta} = 2 X A_{\text{cl}} S \tilde{K}^\top, \quad \frac{\partial V_{\text{wc}}}{\partial W_\theta} = X, \quad (22)$$

(with $\partial V_{\text{wc}}/\partial Q = S$ and the R -block of $\tilde{K} S \tilde{K}^\top$), chained through (12) against the model Jacobians.

The regularity of this route, unlike the minimal- γ SDP (Remark 6), is clean and checkable.

Proposition 2 (Fixed- γ H_∞ regularity). Fix γ above the optimal attenuation and let $X \succeq 0$ be the stabilizing solution

of the game GARE (20), with $R + B_\theta^\top X B_\theta \succ 0$ and the game closed loop A_{cl} Schur. Then X is a locally unique C^1 function of θ : the GARE residual $\mathcal{R}(X, \theta) = 0$ has, at such an X , a linearization in X equal to the closed-loop Stein operator $\delta X \mapsto \delta X - A_{\text{cl}}^\top \delta X A_{\text{cl}}$, which is invertible precisely because $\rho(A_{\text{cl}}) < 1$; the implicit function theorem then gives $X(\theta) \in C^1$ and the closed-form gradient (22) for $V_{\text{wc}}(\theta) = \text{tr}(XW_\theta)$. The dual filter GARE is regular under the analogous conditions, giving $V_{\text{est}}(\theta) \in C^1$.

This is the result the MTDC study of Section 9.4 relies on: it works at a fixed γ , so the delicate minimal- γ rank drop never enters.

The *filter* GARE is the exact dual, giving the estimation half of an output-feedback robust co-design. For the plant (15) with measurement $y_k = C_\theta x_k + D_v v_k$ and estimation-error output $\zeta = L_e x$ ($L_e = M^{1/2}$), the worst-case a-priori error covariance Σ_e solves the control GARE (20) on the *dual* data $(A_\theta^\top, [C_\theta^\top \ L_e^\top], W_\theta, \text{diag}(V, -\gamma^2 I))$ — measurement injection $C_\theta^\top$ and the estimated output $L_e^\top$ playing the roles of B_θ and E_θ — and the worst-case estimation cost is $V_{\text{est}}(\theta) = \text{tr}(M\Sigma_e)$, the robust twin of the Kalman value $\text{tr}(M\Sigma^\varepsilon)$; its gradient is the dual of (22) (contract $2\Sigma_e \tilde{A}_{\text{cl}} \tilde{S}$ against $\partial A_\theta^\top / \partial \theta$, and the C_θ, V blocks of $2\Sigma_e \tilde{A}_{\text{cl}} \tilde{S} \tilde{K}^\top$). This is the form used in Section 9.4: control value $\text{tr}(XW)$ and estimation value $\text{tr}(M\Sigma_e)$ with exact gradients at $O(r_x^3)$, no LMI solve and no rank-deficient dual. The fixed- θ separation into two Riccati is, again, only an inner convenience: the shared θ moves A_θ and hence both halves.

Two remarks connect this to the rest of the framework. First, the weight/noise co-design of Section 5.1 carries over unchanged: the disturbance weights $E_\theta = W_\theta^{1/2}$, $D_v = V_\theta^{1/2}$ and the performance weights in C_z, D_z are again data objects $D(\theta)$, so “how good a sensor to buy” remains a continuous robust co-design variable, priced against J_{des} . Second, the choice between the H_2 /LQG and H_∞ inner values is a modeling decision about the disturbance — average-case stochastic versus worst-case ℓ_2 — that ContEst treats identically at the outer level: both expose an exact gradient from an inner dual, and the outer method of Section 8 is agnostic to which is used.

6 The locally linearized case: MPC and the eKF

Beyond LQG we treat nonlinear dynamics and outputs with additive disturbances, $x_{k+1} = \tilde{f}(x_k, u_k; \theta_f) + w_k$, $y_k = \tilde{h}(x_k; \theta_h) + v_k$, with w_k, v_k zero-mean of covariances Σ_w, Σ_v (possibly functions of θ). The Bayesian filter (4)–(5) is infinite dimensional; we approximate it by an extended Kalman filter (eKF) [5,31], propagating the first two moments. With posterior mean and covariance $x_{k|k}, \Sigma_{k|k}$, the time update is $x_{k+1|k} = \tilde{f}(x_{k|k}, u_k)$, $\Sigma_{k+1|k} = F_k \Sigma_{k|k} F_k^\top + \Sigma_w$, and the measurement update is $x_{k|k} = x_{k|k-1} + L_k(y_k -$

$\bar{h}(x_{k|k-1})$, $\Sigma_{k|k} = \Sigma_{k|k-1} - L_k H_k \Sigma_{k|k-1}$, with $L_k = \Sigma_{k|k-1} H_k^\top (H_k \Sigma_{k|k-1} H_k^\top + \Sigma_v)^{-1}$ and Jacobians $F_k = \partial \bar{f} / \partial x|_{x_{k|k}}$, $H_k = \partial \bar{h} / \partial x|_{x_{k|k-1}}$. The eKF is exact only for affine $\bar{f}$, $\bar{h}$ and Gaussian noise [5].

Quadratic cost in the information state. Take the stage cost quadratic about a reference (set-point) trajectory (s_k^x, s_k^u) , $\ell(x_k, u_k) = \|x_k - s_k^x\|_Q^2 + \|u_k - s_k^u\|_R^2$ —exact for a quadratic regulator, and otherwise the second-order Taylor model of a smooth cost about that trajectory. Lemma 4 with the eKF Gaussian then gives the following.

Lemma 7. *Under the above, $\mathbb{E} \ell(x_k, u_k) = \mathbb{E} \{ \|x_{k|k} - s_k^x\|_Q^2 + \|u_k - s_k^u\|_R^2 + \text{tr}(Q \Sigma_{k|k}) \}$.*

The estimation cost thus appears as $\text{tr}(Q \Sigma_{k|k})$, summed over the horizon—the quantity through which the sensing parameters θ_h act on the objective.

Convex MPC over the information state. Stack the mean and the upper-triangular part of the covariance into the information state $\zeta_k = (x_{k|k}, \text{vech} \Sigma_{k|k}) \in \mathbb{R}^{n_\zeta}$, $n_\zeta = r_x + r_x(r_x + 1)/2$, with lifted dynamics induced by the (prediction-mode) eKF map. Linearizing about the nominal information trajectory yields A_θ^{info} , B_θ^{info} , and the second-stage problem becomes the convex program—the *surrogate* $\hat{J}_c$ that the outer loop actually descends, distinct from both the true stochastic value J_{stoch} of (2) and the realized closed-loop cost J_c evaluated by Monte-Carlo in Section 9 (see the exactness ledger, Table 2),

$$\begin{aligned} \hat{J}_c(\theta) = \min_{\zeta_{1:N}, u_{1:N-1}} \sum_{t=1}^{N-1} \ell(\zeta_t, u_t) + \ell^N(\zeta_N) \\ \text{s.t. } \zeta_{t+1} = A_\theta^{\text{info}} \zeta_t + B_\theta^{\text{info}} u_t, \quad \zeta_1 = \bar{\zeta}, \\ \Sigma_{1|1} \succ 0, \quad c(\zeta_{t+1}, u_t) \leq 0, \end{aligned} \quad (23)$$

where ℓ now carries the $\text{tr}(Q \Sigma_{t|t})$ term of Lemma 7 and $c(\cdot)$ collects state/input constraints.

Design gradients from the dynamics costates. Let λ_t be the multipliers (costates) of the dynamics equalities in (23), returned by the solver. By Theorem 1 the only explicit θ -dependence is through A_θ^{info} , B_θ^{info} , so

$$\frac{\partial \hat{J}_c}{\partial \theta_i} = \sum_t \lambda_t^\top \left(\frac{\partial A_\theta^{\text{info}}}{\partial \theta_i} \zeta_t^* + \frac{\partial B_\theta^{\text{info}}}{\partial \theta_i} u_t^* \right). \quad (24)$$

The factors $\partial A_\theta^{\text{info}} / \partial \theta_i$ and $\partial B_\theta^{\text{info}} / \partial \theta_i$ are Jacobians of the eKF-induced linearization, obtained by automatic differentiation of the *filter map*, never of the optimizer.

Remark 8. Equations (11)–(12) and (24) are one principle in two guises: the inner dual (LMI dual S , or costate λ) contracts against a model Jacobian to give the exact design gradient. This is the computational heart of ContEst.

Table 2

Exactness ledger. J_{stoch} : true stochastic value (2); $\hat{J}_c$: convex surrogate (23); J_c : realized Monte-Carlo closed-loop cost (Section 9). The envelope gradient is exact for the “exact for” column; the last column names the residual gap to J_{stoch} .

regime	inner problem	gradient exact for	gap to true value
LQG (Sec. 5)	SDPs/Riccati	$J_{\text{cont}}, J_{\text{est}}$ = true LQG cost	none
H_∞ fixed γ (Sec. 5.2)	game GARE	$V_{\text{wc}}, V_{\text{est}}$	none (given ℓ_2 model)
nonlinear (Sec. 6)	convex prog. (23)	surrogate $\hat{J}_c$	eKF closure + linearization + CE policy

Remark 9 (Where exactness stops: the nonlinear case). Theorem 1 applies verbatim to the LQG SDPs, which are genuinely convex; in the nonlinear regime it applies to the convex program (23), not to the underlying stochastic control value (2). Three gaps separate the two, and we are explicit about them. (i) *Filtering approximation*: the eKF replaces the exact Bayesian recursion (4)–(5) by its first two moments, exact only for affine $\bar{f}$, $\bar{h}$ and Gaussian noise [5]. (ii) *Non-separability and nonconvexity*: outside LQG the separation of Section 5 fails, and $J_{\text{stoch}}(\theta)$ is in general non-convex and possibly nonsmooth in θ , so a single stationary point need not be global—hence the multi-start outer loop of Section 9. (iii) *Linearization*: A_θ^{info} , B_θ^{info} are taken about a nominal information trajectory, so (24) is the *exact* gradient of the convex surrogate but an *approximate* gradient of the true value; the surrogate is what the outer method actually descends. Assumption 1 is therefore imposed on (23), and the gradient check of Section 9 confirms it holds for the surrogate. Making the certainty-equivalent and linearization gaps quantitative—error bounds relating the surrogate stationary point to a stationary point of (2)—is left as future work; a sharper information state than the eKF (e.g. the eKF–Koopman lift [31]) narrows gap (i).

Table 2 is the resulting *exactness ledger*: it separates the three objects the analysis involves—the true stochastic value J_{stoch} of (2), the convex surrogate $\hat{J}_c$ of (23), and the realized closed-loop cost J_c that Section 9 evaluates by Monte-Carlo—and states, per regime, what the envelope gradient (7)/(24) is *exact* for and what gap remains to the true value. In the LQG and fixed- γ H_∞ regimes the inner value is the true stationary cost and the gradient is exact (Propositions 1, 2); in the nonlinear regime the gradient is exact for the surrogate $\hat{J}_c$ only, and the Monte-Carlo evaluation of Section 9 does theoretical work by empirically bounding the surrogate gap—each optimized design is shown to improve the *realized* closed-loop cost, not merely the surrogate.

7 Sensor allocation: discrete selection and sparse relaxation

A recurring design decision is *which* sensors to deploy. With r_y candidate sensors, binary indicators $b_j \in \{0, 1\}$ (sensor j contributing its row C_j and noise variance $\sigma_{v,j}^2$ when $b_j = 1$) and per-sensor costs c_j make the co-design $\min_{\theta, b \in \{0,1\}^{r_y}} J_{\text{stoch}}(\theta, b) + c^\top b$ a mixed-integer (SDP) program, whose exact solution is combinatorial (NP-hard, 2^{r_y} subsets) [25,40]. Standard tractable remedies—greedy selection with submodularity guarantees [40,42], the convex Boolean relaxation $b_j \in [0, 1]$ [25], and sparsity-promoting ℓ_1 [15,34,14]—are all compatible with ContEst; we adopt the ℓ_1 route, which fits it with no change to the gradient machinery.

Replace the indicators by continuous gains $\alpha_j \geq 0$ scaling each sensor’s contribution, $C(\alpha) = \text{diag}(\alpha) \bar{C}_{\text{base}}$ ($\alpha_j = 0$ removing sensor j), so α joins θ through C_θ , and put the sparsity penalty in the *design cost*,

$$J_{\text{des}}(\theta) = J_{\text{hw}}(\theta) + \lambda_s \|\alpha\|_1, \quad \alpha \geq 0, \quad (25)$$

with J_{hw} the smooth design penalties and $\lambda_s > 0$ trading estimation quality against the number of active sensors. Because the ℓ_1 term sits in J_{des} and not the inner problem, the envelope gradients (12), (24) are untouched. Its one kink is removed by the standard epigraph lift [12]: substitute $\lambda_s \|\alpha\|_1 \mapsto \lambda_s \mathbf{1}^\top t$ and append $\alpha \preceq t$ ($\alpha \geq 0$), which is smooth (linear in the slacks t) over a polyhedron, so the (L-)BFGS of Section 8 keeps its convergence guarantees; equivalently one takes a proximal (soft-threshold) step on $\lambda_s \|\alpha\|_1$. When a binary deployment is required, the converged α^* is thresholded (optionally with a fixed-sensor resolve, or reweighted- ℓ_1 iterations [13]). ContEst thus treats sensor allocation and continuous sensor tuning uniformly; the MTDC study of Section 9.4 exercises this allocation on the robust H_∞ path, pruning 60% of the DC-voltage sensors at no estimation cost.

8 Outer optimization

Problem 1 is solved by a projected quasi-Newton method on θ over the box $\Theta = [\theta_{\text{lb}}, \theta_{\text{ub}}]$. Each outer iteration: (1) solves the inner convex problem ((9)/(10) or (23))—one solve, or one per scenario when several are averaged (Remark 2); (2) reads off its duals (S or λ); (3) assembles the design gradient via (12) or (24); (4) adds ∇J_{des} , including the ℓ_1 subgradient of (25); and (5) takes a quasi-Newton (e.g. L-BFGS) step with projection onto Θ . Because the gradients are exact—neither finite-difference nor approximate implicit derivatives—the outer iteration retains the fast local convergence of quasi-Newton methods, and the dominant per-iteration cost is the inner convex solves, which scale polynomially.

Stationarity. When the objective is smooth—the LQG and fixed- γ H_∞ regimes, where Propositions 1 and 2 make

$J \in C^1$, and the nonlinear surrogate (23) likewise—every accumulation point of the projected quasi-Newton iterates is a stationary point of J over Θ [8]. The one nonsmooth term, the sparsity penalty $\lambda_s \|\alpha\|_1$ of (25), is handled without forfeiting this guarantee by the epigraph lift of Section 7 (equivalently, a proximal-gradient step on the composite objective [8]).

9 Simulation studies

We evaluate ContEst on three nonlinear co-design problems spanning very different physics: spacecraft attitude determination and control (Section 9.1), a binary distillation column (Section 9.2), and PLL/DSE estimator tuning in a low-inertia grid (Section 9.3), followed by a *large-scale, linear* power-system study on the steady-state *robust* (H_∞) inner problem (multi-terminal HVDC droop coordination, Section 9.4) and a complexity analysis of the two inner problems under interior-point solvers (Section 9.5). Each of these three is genuinely nonlinear, so all use the eKF-MPC second stage of Section 6: the convex program (23) is solved with the Clarabel interior-point solver, and the design gradient is assembled from the dynamics costates through (24). The outer loop is the projected quasi-Newton method of Section 8 (L-BFGS) over the box Θ , run from 5 initial points—the baseline design θ_{nom} plus four uniform random restarts in Θ —keeping the best minimizer. For the ADCS and PLL studies all five starts *converge to the same minimizer*, so the reported design is the global optimum over Θ to numerical tolerance; the distillation study is genuinely multimodal (the restarts resolve three distinct minima), and we report the best, the restarts escaping the local optimum the baseline start alone would return. The envelope-theorem gradients are exact, so we do not verify them per study; as a one-time sanity check we confirmed, across all studies below, that the analytic gradient matches central finite differences to a relative error below 10^{-4} (best case 1.6×10^{-6}). Code reproducing every study is publicly available.¹

In every study the design splits as $\theta = (\theta_f, \theta_h)$, where θ_f enters the dynamics $\bar{f}$ (actuation / plant tuning, the classical CCD axis) and θ_h is the estimation axis that makes the problem ContEst. Each axis is posed so that its optimum is a genuine *interior* trade—a scarce resource allocated under a budget, or a saturating benefit balanced against a provisioning cost—rather than a monotone “more is better” knob pinned at a bound. The reported costs are *Monte-Carlo averages of the realized closed-loop cost*: for each design we draw independent initial states and process/measurement noise, run the true nonlinear plant forward under a receding-horizon controller with the eKF processing the noisy measurements, and average the incurred cost over the sampled trajectories. We decompose it into its two physical parts, both evaluated

¹ <https://github.com/anonymous/ContEst> (placeholder repository; the final URL will be provided on acceptance).

on the *true* state x_t ,

$$\begin{aligned} J_{\text{det}} &= \sum_t (\|x_t - s_t^x\|_Q^2 + \|u_t - s_t^u\|_R^2), \\ J_{\text{est}} &= \sum_t \|x_t - \hat{x}_{t|t}\|_Q^2, \end{aligned} \quad (26)$$

the (deterministic) control cost and the realized estimation error against the eKF mean $\hat{x}_{t|t}$ (whose expectation is $\sum_t \text{tr}(Q\Sigma_{t|t})$), with $J_c = J_{\text{det}} + J_{\text{est}}$ and reported total $J_{\text{tot}} = J_c + J_{\text{des}}$ (single scenario, $M = 1$; Remark 2). The *baseline* design is driven by a certainty-equivalence controller that uses *only the eKF mean* (it plans on the mean dynamics and is blind to the covariance), whereas the ConTEst design is driven by the covariance-aware information-state MPC of Section 6; the reported reduction therefore reflects both the re-designed parameters and the estimation-aware control they enable. (The distillation study, whose many-stage information-state MPC is expensive to simulate in closed loop, instead reports the open-loop surrogate $\hat{J}_c$ of (23) directly, with $J_{\text{est}} = \sum_t \text{tr}(Q\Sigma_{t|t})$ from the eKF covariance rollout.) The design optimization itself is unchanged—multi-start L-BFGS on the envelope-gradient objective. As anticipated by the exactness ledger (Table 2), the surrogate $\hat{J}_c$ is an *open-loop* planned cost and is therefore optimistic relative to the noise-driven realized cost J_c ; what matters for design is that minimizing the surrogate also lowers the realized cost, which the Monte-Carlo evaluation confirms in every study, with the design ranking preserved.

9.1 Spacecraft attitude determination and control (ADCS)

Significance. Pointing accuracy on every space mission is ultimately set by both the attitude *estimator* (the sensor suite: star trackers, gyros) and the reaction-wheel authority. On a spacecraft these compete for a shared, tightly constrained *power/mass budget*: a heavier, higher-power star tracker or gyro reports lower noise, and a larger wheel delivers more torque, but every watt and kilogram spent on one is unavailable to the others. The benefit of each saturates, so the design question is not “how good a sensor” but how to *split one budget* across sensing and actuation—an allocation classical control co-design cannot pose because it has no sensing axis.

Formulation. A rigid satellite with asymmetric inertia $J = \text{diag}(4, 6, 5) \text{ kg m}^2$ regulates attitude and rate to zero. The state $x = [\phi, \theta_a, \psi, \omega_x, \omega_y, \omega_z]$ ($r_x = 6$) uses small-angle attitude and body rates; the input u is three reaction-wheel torques ($r_u = 3$); the outputs are a star tracker (attitude) and a rate gyro ($r_y = 6$). The genuine nonlinearity is the gyroscopic coupling $\omega \times (J\omega)$, which vanishes at the origin so $\bar{f}(0, 0; \theta) = 0$ for all θ ; integration is at 0.1 s over $N = 15$. The three design parameters—reaction-wheel authority e_{rw} (in $\bar{f}$) and the star-tracker and rate-gyro precisions $\alpha_{\text{st}}, \alpha_{\text{gyro}}$ (entering the measurement covariance $V(\theta)$)—draw on a single resource budget. Attitude error is weighted four times

Table 3

Spacecraft ADCS: baseline (uniform budget) vs. co-designed allocation of the shared power/mass budget $e_{\text{rw}} + \alpha_{\text{st}} + \alpha_{\text{gyro}} \approx B = 3$.

parameter	axis	baseline	optimal
e_{rw}	f (reaction wheels)	1.000	1.186
α_{st}	V (star tracker)	1.000	1.642
α_{gyro}	V (rate gyro)	1.000	0.777

more heavily than rate, so the star tracker is intrinsically more valuable than the gyro.

State-space model. With $\Delta t = 0.1 \text{ s}$ the system (1) reads $x_{k+1} = \bar{f}(x_k, u_k; \theta) + w_k$, $y_k = \bar{h}(x_k) + v_k$ with

$$(\phi, \theta_a, \psi)^+ = (\phi, \theta_a, \psi) + \Delta t (\omega_x, \omega_y, \omega_z), \quad (27a)$$

$$\omega^+ = \omega + \Delta t J^{-1}(e_{\text{rw}} u - \omega \times (J\omega)), \quad (27b)$$

$$\bar{h} = [(\phi, \theta_a, \psi), (\omega_x, \omega_y, \omega_z)], \quad (27c)$$

the gyroscopic term $\omega \times (J\omega)$ the nonlinearity (zero at $\omega = 0$), and θ -dependent sensor noise $V(\theta) = \text{diag}(v_0/\alpha_{\text{st}}^2 I_3, v_0/\alpha_{\text{gyro}}^2 I_3)$ (better precision $\Rightarrow$ lower variance, saturating). *Noises:* $w_k \sim \mathcal{N}(0, \Sigma_w)$ with $\Sigma_w = \text{diag}(10^{-6} I_3, 10^{-4} I_3)$, base sensor variance $v_0 = 0.20$, prior $\Sigma_0 = 2 \times 10^{-2} I_6$. *Cost:* $Q = \text{diag}(200 I_3, 50 I_3)$, $R = I_3$, with $|\tau_i| \leq 0.8 \text{ N m}$. *Design cost:* a shared budget $c_b(e_{\text{rw}} + \alpha_{\text{st}} + \alpha_{\text{gyro}} - B)^2$ ($c_b = 30$, $B = 3$), with box $e_{\text{rw}} \in [0.3, 3]$, $\alpha_{\text{st}}, \alpha_{\text{gyro}} \in [0.3, 5]$, and baseline $\theta_{\text{nom}} = 1$ (budget spent uniformly).

Results. All five starts agreed. The co-design (Table 3) re-allocates the budget: it upgrades the star tracker most ($\alpha_{\text{st}} : 1 \rightarrow 1.64$), trims the lightly weighted rate gyro ($\alpha_{\text{gyro}} : 1 \rightarrow 0.78$), and modestly raises wheel authority ($e_{\text{rw}} : 1 \rightarrow 1.19$)—none pinned at a bound. This cuts the estimation cost J_{est} by 23.8% ($173.1 \rightarrow 132.0$) and the control cost J_{det} by 4.0% ($448.3 \rightarrow 430.5$), for a 9.5% reduction in $J_c = J_{\text{det}} + J_{\text{est}}$ ($621.4 \rightarrow 562.5$) and a 7.7% reduction in the total J_{tot} ($621.4 \rightarrow 573.5$).

Comments. With a single shared budget the axes genuinely compete, and the optimizer spends it where the cost weights make it count—attitude sensing (the star tracker) over rate sensing (the gyro), with a little left for wheel authority. The interior split is exactly the effort allocation *across* the sensing and actuation axes that a separate, control-only design cannot represent. A sensor-selection method [25,40] cannot pose this trade because it has no actuation axis to spend the shared budget against; plant–controller co-design cannot pose it because it has no sensing axis at all.

9.2 Binary distillation column

Significance. Distillation is the workhorse separation of the process industries and its control is dominated by two classical layout decisions: on *which* tray to introduce the feed, and

on *which* tray to place the inferential (temperature) sensor that stands in for an unmeasured composition profile. Both are *placement* choices with well-known interior optima—a feed stage too high or too low wastes separation, and the most informative temperature tray is the steep mid-column stage, not an end tray. The feed locations shape the plant $\bar{f}$ while the sensor locations shape the output map $\bar{h}$, so this is a ContEst problem in which *neither* axis is a gain: every parameter moves a discrete-in-spirit location on a continuous scale.

Formulation. A twelve-stage column (stage 1 reboiler ... stage 12 condenser) separates a light/heavy mixture; the state $x = [\Delta x_1, \dots, \Delta x_{12}]$ ($r_x = 12$) collects stage light-component mole-fraction deviations about the nominal profile, in deviation form so $\bar{f}(0, 0; \theta) = 0$ for every θ . The nonlinearity is genuine and enters twice: the vapour-liquid equilibrium $y(x) = \alpha x / (1 + (\alpha - 1)x)$ ($\alpha = 2.5$) and the bubble-point temperature $T(x) = T_{B0} - \Delta T_B y(x)$. The single input is the reboiler boil-up deviation $u = \Delta V$ ($r_u = 1$), limited to $|\Delta V| \leq 1.5$; the measurements ($r_y = 3$) are three Gaussian-weighted stage temperatures read at the sensor locations. Integration is at 0.05 s over $N = 12$. The design carries $N_f = 3$ feed-stage locations $\theta_f = (f_1, f_2, f_3)$ and $N_s = 3$ temperature-sensor locations $\theta_h = (s_1, s_2, s_3)$, all in [2, 11]; the total feed is distributed over trays by a *sum* of normalised Gaussians centred at the f_j , so the cumulative feed above each stage sets that stage's internal liquid traffic $L_i(\theta_f)$ and the feed placement nonlinearly reshapes the whole convective coupling of the column. The upset is a feed-composition disturbance that peaks mid-column, and the stage weights Q are heaviest there. Because temperature is the *only* measurement, where the three sensors sit sets how well the entire composition profile is reconstructed, and they must spread to cover it rather than pile onto one tray. Placement is nearly free to relocate, so J_{des} carries only a mild regularisation $\lambda_{\text{des}} \|\theta - \theta_{\text{nom}}\|^2$ about the naive default θ_{nom} with feeds at (3, 4.5, 6) and sensors at (2, 3.5, 5) (feeds below centre, sensors low near the bottoms product)—the win is driven by the column physics, not by the design penalty.

State-space model. With equilibrium and temperature maps $y(x) = \alpha x / (1 + (\alpha - 1)x)$, $T(x) = T_{B0} - \Delta T_B y(x)$, equilibrium deviations $\Delta y_i = y(x_i^{\text{ss}} + \Delta x_i) - y_i^{\text{ss}}$ and $\Delta t = 0.05$ s, the system (1) reads $x_{k+1} = \bar{f}(x_k, u_k; \theta) + w_k$, $y_k = \bar{h}(x_k; \theta) + v_k$ with, for $i = 1, \dots, 12$ and $j = 1, \dots, 3$,

$$\Delta x_i^+ = \Delta x_i + \frac{\Delta t}{M} (\text{liq}_i + \text{vap}_i), \quad (28a)$$

$$\text{liq}_i = L_i(\theta_f) (\Delta x_{i+1} - \Delta x_i), \quad (28b)$$

$$\text{vap}_i = (V + u) (\Delta y_{i-1} - \Delta y_i) + u (y_{i-1}^{\text{ss}} - y_i^{\text{ss}}), \quad (28c)$$

$$\bar{h}_j = \sum_{i=1}^{12} \hat{w}_i(s_j) (T(x_i^{\text{ss}} + \Delta x_i) - T_i^{\text{ss}}), \quad (28d)$$

boundary terms $\Delta x_{13} = \Delta y_0 = y_0^{\text{ss}} = 0$, feed weights $w_k(\theta_f) \propto \sum_{j=1}^3 e^{-(k-f_j)^2/2\sigma_f^2}$ (normalized) setting the internal liquid traffic $L_i(\theta_f) = L_{\text{ref}} + F \sum_{k>i} w_k$, and sensor weights $\hat{w}_i(s_j) \propto e^{-(i-s_j)^2/2\sigma_{\text{sens}}^2}$ (normalized)—the maps carrying the design coupling. Constants $\alpha = 2.5$,

Table 4

Binary distillation: naive baseline vs. co-designed placements. All six parameters are continuous tray locations (not gains); baseline θ_{nom} places feeds at (3, 4.5, 6) and sensors at (2, 3.5, 5).

parameter	axis	baseline	optimal
f_1	f (feed stage)	3.000	2.479
f_2	f (feed stage)	4.500	3.584
f_3	f (feed stage)	6.000	9.018
s_1	h (temp. sensor)	2.000	2.972
s_2	h (temp. sensor)	3.500	5.428
s_3	h (temp. sensor)	5.000	8.010

$L_{\text{ref}} = 2$, $V = 2.5$, $F = 2.5$, $M = 1$, $\sigma_f = \sigma_{\text{sens}} = 1.2$, $T_{B0} = 380$ K, $\Delta T_B = 40$ K. *Noises:* $\Sigma_w = 2 \times 10^{-5} I_{12}$, $\Sigma_v = 0.6 I_3$ (K²), prior $\Sigma_0 = 10^{-2} I_{12}$. *Cost:* $Q_{ii} = 20 + 30 e^{-(i-6.5)^2/2(12/5)^2}$ (heaviest mid-column), $R = 2$, $|u| \leq 1.5$, with a mild placement regularization $\lambda_{\text{des}} \|\theta - \theta_{\text{nom}}\|^2$.

Results. The design landscape is multimodal: the five-start L-BFGS resolves three distinct minima, and the baseline start alone converges to an inferior one, so the random restarts are what recover the reported optimum. ContEst spreads the feed streams—two settle lower toward the reboiler ($f_{1,2} : 3.0, 4.5 \rightarrow 2.5, 3.6$) while the third climbs high up the column ($f_3 : 6.0 \rightarrow 9.0$)—and fans the three temperature sensors out of the bottoms region across the steep mid-and-upper trays ($s : (2.0, 3.5, 5.0) \rightarrow (3.0, 5.4, 8.0)$), covering the composition profile rather than clustering low (Table 4). This cuts the estimation cost J_{est} by 22.8% ($12.8 \rightarrow 9.9$) and the control cost J_{det} by 1.9%, for a 10.2% reduction in J_c and—after the small placement penalty—a 6.5% reduction in the total J_{tot} ($32.3 \rightarrow 30.2$).

Comments. This study stresses ContEst on the *placement* form of the sensing axis—three movable inferential sensors and three feed streams—on a nonconvex, multimodal objective. Both physical effects that make distillation layout hard—the interior-optimal feed distribution and the interior-optimal sensor trays—are recovered jointly by the same envelope gradient, and the gain is estimation-led (J_{est} down 22.8% against a nearly flat J_{det}), reflecting that with temperature the only measurement the reconstruction quality dominates the achievable regulation. It also illustrates why co-design matters even without a hard sensor budget: a control-only design would inherit the naive low sensor trays and forfeit the estimation-cost reduction, while the baseline start alone would stall at a local minimum. A discrete sensor-selection method cannot represent this design either, because the feed and sensor trays are *continuous* locations whose interior optima fall between integer stages—the coupling of $\bar{f}$ and $\bar{h}$ through the column physics is invisible to menu enumeration.

9.3 Estimator co-design: PLL/DSE tuning in a low-inertia grid

Significance. The studies above move the sensing axis θ_h through the output map—sensor precision or placement, i.e. *what* or *where* one senses. This study co-designs the *estimator itself*—the per-inverter dynamic state estimation (DSE). Every grid-following (GFL) inverter estimates grid frequency and phase with a phase-locked loop (PLL), whose bandwidth is a classic bias–variance knob: a narrow-band PLL is smooth but sluggish and lags real frequency excursions, while a wide-band PLL tracks fast but passes more measurement noise into the frequency estimate—and in low-inertia, inverter-dominated grids this tuning is decisive for stability and for fast frequency response [43]. The PLL bandwidth is therefore an *estimator* design variable that classical control co-design cannot represent and sensor placement cannot capture. It is also the first study to exercise the noise-level extension of Section 5.1: the PLL bandwidth enters both the dynamics $\bar{f}$ (the PLL tracking state) *and* the measurement covariance $V(\theta)$ (the frequency-readout noise), so θ_h shapes V rather than $\bar{h}$.

Formulation. Three GFL-inverter buses form a low-inertia chain. The state $x = [\Delta\delta_{1:3}, \Delta\omega_{1:3}, \Delta\hat{\omega}_{1:3}^p]$ ($r_x = 9$) stacks bus angle deviations, frequency deviations, and the *per-bus PLL frequency estimate* $\Delta\hat{\omega}_i^p$; the input $u = [\Delta P_{1:3}]$ is the inverter damping power ($r_u = 3$). Each bus is measured by a phasor measurement unit (PMU) angle channel and its PLL frequency channel ($r_y = 6$). The genuine nonlinearity is the synchronizing power $\propto \sin(\delta_{ij}^{\text{eq}} + \Delta\delta_i - \Delta\delta_j)$, in deviation form so $\bar{f}(0, 0; \theta) = 0$ for all θ ; integration is at 0.05 s over $N = 12$. The design parameters are the per-bus inverter damping effectiveness $\theta_f = (e_1, e_2, e_3) \in [0.3, 3]^3$ and the per-bus PLL bandwidths $\theta_h = (b_1, b_2, b_3) \in [2, 20]$ rad/s. The disturbance is concentrated at bus 1 (large post-fault kick and prior), while buses 2,3 are quiet, so the design must trade tracking speed against readout noise *per bus*.

State-space model. With $\Delta t = 0.05$ s the system reads $x_{k+1} = \bar{f}(x_k, u_k; \theta) + w_k$, $y_k = \bar{h}(x_k) + v_k$ with, for $i = 1, 2, 3$,

$$\Delta\delta_i^+ = \Delta\delta_i + \Delta t \Delta\omega_i, \quad (29a)$$

$$\Delta\omega_i^+ = \Delta\omega_i + \frac{\Delta t}{2H_i} (-P_i^e(\Delta\delta) - D_i\Delta\omega_i + e_i u_i), \quad (29b)$$

$$\Delta\hat{\omega}_i^{p,+} = \Delta\hat{\omega}_i^p + \Delta t b_i (\Delta\omega_i - \Delta\hat{\omega}_i^p), \quad (29c)$$

$$\bar{h} = [\Delta\delta_i; \Delta\hat{\omega}_i^p]_{i=1}^3, \quad (29d)$$

where $P_i^e(\Delta\delta) = \sum_{j \neq i} K_{ij} [\sin(\Delta\delta_{ij}^{\text{eq}} + \Delta\delta_i - \Delta\delta_j) - \sin \Delta\delta_{ij}^{\text{eq}}]$ is the synchronizing power ($H_i, D_i, \delta_i^{\text{eq}}$ the machine inertias, dampings and equilibrium angles) and the third line is the first-order PLL tracking the true frequency with bandwidth b_i . *Noise:* $\Sigma_w = \text{blkdiag}(2 \times 10^{-5} I_3, 5 \times 10^{-4} I_3, 5 \times 10^{-4} I_3)$; prior Σ_0 large at bus 1 (3×10^{-2}), small elsewhere (6×10^{-3}). *Cost:* $Q = \text{blkdiag}(15 I_3, 120 I_3, 0_3)$

(weighting the *true* angle and, heavily, the true frequency; PLL states unweighted), $R = I_3$, $|u_i| \leq 1.5$. *Design cost:* $J_{\text{des}} = \lambda_e \sum_i (e_i - 1)^2 + \lambda_b \sum_i (b_i - b_{\text{nom}})^2$, $\lambda_e = 0.5$, $\lambda_b = 0.02$, $b_{\text{nom}} = 6$. The design coupling is that the PLL bandwidth b_i enters *both* $\bar{f}$ (the tracking line above) and the measurement covariance, which is *design-dependent*,

$$V(\theta) = \text{blkdiag}(\sigma_\delta^2 I_3, \text{diag}_i \sigma_\omega^2 (1 + \kappa b_i)), \quad (30)$$

$\sigma_\delta^2 = 4 \times 10^{-3}$, $\sigma_\omega^2 = 2 \times 10^{-3}$, $\kappa = 0.30$: a wider-band PLL tracks faster but yields a noisier frequency readout—the bias–variance knob θ_h resolves.

Results. Here the envelope-theorem gradient flows through the *design-dependent covariance* $V(\theta)$ —the numerical exercise of the noise-level extension of Section 5.1. All five starts converged to the same minimizer, so the reported design is the global optimum over Θ to numerical tolerance. The naive uniform PLL ($b = 6$) is too sluggish; ContEst raises every bandwidth but tunes each to its local signal-to-noise, most aggressively at the disturbed bus ($b_1: 6 \rightarrow 13.7$) and moderately at the quiet buses ($b_{2,3}: 6 \rightarrow 10.6$)—and *none* reaches the upper bound of 20, because beyond the optimum the added readout noise in $V(\theta)$ outweighs faster tracking (Table 5). It simultaneously raises the inverter damping authority ($e_1 \rightarrow 3.0$, $e_{2,3} \rightarrow 1.95, 2.19$). This cuts the estimation cost J_{est} by 20.9% ($26.6 \rightarrow 21.0$) and the control cost J_{det} by 40.6% ($151.9 \rightarrow 90.2$), for a 37.7% reduction in $J_c = J_{\text{det}} + J_{\text{est}}$ ($178.5 \rightarrow 111.2$) and a 34.8% reduction in the total J_{tot} ($178.5 \rightarrow 116.4$).

Comments. This is the most vivid demonstration that the *estimator* is a first-class design variable in ContEst. The tuned quantity is neither a sensor location nor a static gate but the bandwidth of a *dynamic* estimator embedded in every inverter, and it acts on the objective through the measurement covariance $V(\theta)$ —a channel absent from every prior study and from control-only co-design. That the optimal bandwidths sit strictly inside the box is the signature of the bias–variance trade-off the framework now resolves automatically, allocating the most estimator bandwidth exactly where the dynamics are fastest and least certain. No sensor-placement method can tune this variable, because it tunes an estimator *state* in $\bar{f}$ and the noise level $V(\theta)$ rather than *which* sensor to read.

Beyond the naive baseline: a competent sequential pipeline. A practitioner would not use the uniform baseline but a *sequential* pipeline—tuning the estimator θ_h first for a pure estimation criterion (the eKF covariance rollout $\sum_t \text{tr}(Q\Sigma_t)$) and then the control θ_f with θ_h frozen, optionally *alternating* the two. We ran both against the joint optimum; here the designs nearly coincide (the joint optimum improves the surrogate it descends by only 0.3% over the sequential design and ties it within Monte-Carlo noise on the realized cost). This study is, in practice, close to *separable*: the estimation-first tuning already sits near the joint optimum. Two points follow. First, the large reductions reported above are against

Table 5

Low-inertia grid (PLL/DSE): baseline vs. co-designed parameters. The PLL bandwidths b_i enter both $\bar{f}$ and the measurement covariance $V(\theta)$.

parameter	axis	baseline	optimal
e_1	f (inverter damping)	1.000	3.000
e_2	f (inverter damping)	1.000	1.951
e_3	f (inverter damping)	1.000	2.191
b_1	h, V (PLL bandwidth)	6.000	13.718
b_2	h, V (PLL bandwidth)	6.000	10.567
b_3	h, V (PLL bandwidth)	6.000	10.575

the *naive* design that deployed systems actually carry, the relevant baseline. Second—the structural advantage—ContEst does not need to *know* in advance whether a problem separates: one joint pass recovers the competent sequential design when it is near-optimal and captures the coupling where it is strong (the general non-separability of Section 5), without committing to a place-then-control ordering that can, in more strongly coupled plants, strand the design at a non-stationary point.

Finally, because the PLL bandwidths and damping gains are *software-settable* and the useful allocation is operating-point dependent (the localized disturbance fixes *which* bus is fastest and least certain), the low per-solve cost of ContEst—one convex solve plus the envelope contraction of Section 9.5—makes it natural to re-run as a slow, event-triggered design layer that re-provisions set-points as the operating point shifts, complementary to the online input adaptation of dual/adaptive control.

9.4 Multi-terminal HVDC droop coordination

The three studies above are nonlinear and use the eKF–MPC inner problem. We close with a *large-scale, linear* power-system study, and use it to exercise the *robust output-feedback* (H_∞) inner problem of Section 5.2 in place of the H_2 /LQG one. The device–network model is linearized about an operating point and discretized by an explicit Euler step,

$$\begin{aligned} x_{k+1} &= A(\theta) x_k + B u_k + E w_k, & y_k &= C x_k + D_v v_k, \\ z_k &= [Q^{1/2} x_k; R^{1/2} u_k], \end{aligned} \quad (31)$$

with $A = I + \Delta t A_c$, $B = \Delta t B_c$, disturbance channel $E = W^{1/2}$ (renewable infeed), and a measurement y that telemeters only the *DC-bus voltages* (C selects the ΔV_i states, $r_y = 25$, noise $D_v D_v^\top = V = 10^{-3} I$)—the converter *currents* are unmeasured and must be estimated. This is therefore a genuine output-feedback co-design. Fixing the attenuation level $\gamma^2 = 8$, the inner value splits, as in the LQG separation, into a worst-case control and a worst-case estimation cost,

$$J_c(\theta) = \underbrace{\text{tr}(XW)}_{J_{\text{det}}(\theta)} + \underbrace{\text{tr}(M\Sigma_e)}_{J_{\text{est}}(\theta)}, \quad (32)$$

with X from the control game Riccati (20) and Σ_e from its dual filter GARE (estimation-error weight $M = Q$); the design gradient is the sum of the two closed-form contractions (22) and its dual, chained with the Jacobian of $A(\theta)$ —no differentiation through the solver. Although the fixed- θ game separates into two Riccati, the droop θ enters $A(\theta)$ and hence moves *both* X and Σ_e , so J_{det} and J_{est} vary over a common θ and the co-design does not separate. We take the $O(r_x^3)$ game-Riccati route (Section 5.2); the SDP form (16)–(17) is avoided because, at the interior γ optimum it minimizes over, its LMIs are rank-deficient and their duals ill-conditioned (Section 9.5). The outer L-BFGS loop of Section 8 runs from 5 starts (θ_{nom} plus four random restarts). The design gradient here sums the control and the *dual filter* envelope contractions.

Significance — why H_∞ fits here. The failure mode of a droop grid is a *resonant peak*: the converter lag makes excessive droop provoke a lightly damped inter-terminal DC-voltage oscillation, and a renewable-infeed step that lands near that resonance is amplified far beyond its RMS level. It is this worst-case amplification, not the noise-averaged variance, that a secure design must bound—exactly the quantity H_∞ prices, while H_2 averages it away. Both halves inherit the resonance: a near-oscillatory plant is both harder to regulate and harder to *estimate* under a worst-case disturbance, so J_{det} and J_{est} are each sharply *U-shaped* in the droop. A naive design chasing tight regulation over-provisions droop and lands on that wall—the baseline here—so ContEst must *lower* the gains to their interior robust optimum.

State-space model. $n_t = 25$ terminals on a meshed DC grid ($r_x = 50$), each with state $[\Delta V_i, \Delta I_i]$ and, for $\Delta t = 5$ ms,

$$C_i \Delta \dot{V}_i = -\sum_{j \sim i} G_{ij} (\Delta V_i - \Delta V_j) + \Delta I_i + \text{infeed}, \quad (33a)$$

$$\tau_i \Delta \dot{I}_i = -\Delta I_i - (k_0 \theta_i) \Delta V_i + u_i, \quad (33b)$$

$$y_{k,i} = \Delta V_i + v_{k,i}, \quad v_k \sim \mathcal{N}(0, 10^{-3} I_{25}), \quad (33c)$$

so the droop gain $k_0 \theta_i$ ($k_0 = 12$) enters A through the converter-current feedback with lag τ_i . A heterogeneous renewable-infeed disturbance (σ_{inf} per node) drives the DC nodes; Q penalizes ΔV , $R = 2I$, estimation-error weight $M = Q$; small converter-stress cost $J_{\text{des}} = c_k \sum_i \theta_i$, $c_k = 5 \times 10^{-3}$; box $\theta \in [0.5, 4]^{25}$ and naive over-provisioned baseline $\theta_{\text{nom}} = 3$. At this leverage the interior robust optimum is near $\theta \approx 1$ with $J_c(\theta=3) \approx 1.7 J_c(\theta=1)$; only the DC-bus voltages ΔV_i are measured, so the converter currents must be estimated.

Results. All five starts agree, so the optimum is numerically global. Consistent with the *U-shaped* J_c , ContEst *lowers* the over-provisioned droop to an interior, disturbance-dependent profile ($\theta \in [0.66, 0.92]$, none at a bound; noisier terminals keep slightly higher droop), pulling the terminals off the resonance wall. Both halves improve: the worst-case estimation cost J_{est} falls 42.6% ($1.08 \rightarrow 0.62$) and the worst-case

control cost J_{det} falls 39.7% ($2.78 \rightarrow 1.68$), for a 40.5% reduction in J_c ($3.86 \rightarrow 2.30$) and, with the reduced converter stress, a 43.4% reduction in J_{tot} ($4.24 \rightarrow 2.40$). The single control-side design variable improves the estimation half purely by reshaping the shared dynamics $A(\theta)$ —the design-level coupling that a place-then-control pipeline cannot exploit.

Adding a sparse sensing axis (ℓ_1 allocation). The study above telemeters all 25 DC-bus voltages. But the meshed ring couples neighbouring bus voltages, so an unsensed terminal can be reconstructed by the H_∞ filter from its neighbours—which voltages to telemeter is itself a design decision, and the natural vehicle for it is the discrete sensor-allocation relaxation of Section 7, which the studies so far have described but not exercised. We give each measurement row a gain $\alpha_i \geq 0$, $C(\alpha) = \text{diag}(\alpha) C$ (so $\alpha_i = 0$ removes sensor i and larger α_i raises its SNR at fixed channel noise), append α to the design, and add the sparsity penalty $\lambda_s \|\alpha\|_1$ to J_{des} (25). Since $\alpha \geq 0$ the epigraph lift of Section 7 collapses to the smooth linear term $\lambda_s \mathbf{1}^\top \alpha$, and α enters only C , whose envelope gradient (19) is already in hand—so no machinery changes. This promotes MTDC to a genuine two-axis co-design: the control droop $\theta_f = k$ (in A) and the sensing gains $\theta_h = \alpha$ (in C), optimised jointly on the same robust inner value.

Sweeping λ_s traces a sparsity/performance frontier (Table 6). Across the co-designed rows the control cost J_{det} holds at its robust optimum (≈ 1.67), so the frontier isolates the price of sparsity: the worst-case estimation cost degrades *gracefully* as sensors are pruned, because the network coupling makes most voltage telemetry redundant—12 sensors already leave J_{est} at the 25-sensor *naive*-baseline level, and even 6 sensors cost only $\sim 17\%$. At $\lambda_s = 0.06$ the joint co-design keeps 10 of 25 sensors; because the sensor on/off combinatorics make the landscape multimodal (unlike the droop-only study), we run the multi-start of Section 8 and report the best of five. Thresholding that solution to its 10-sensor integer menu and re-solving the droop gives $J_{\text{est}} = 0.72$ and $J_{\text{det}} = 1.67$ — J_c down 38.2% against the full-telemetry baseline while *removing 60% of the voltage sensors*, with estimation in fact *better* than with all 25 (the pruned sensors’ budget is spent on the survivors). This is the discrete allocation of Section 7 in action on the robust path: one envelope gradient tunes the droop and prunes the sensor set at once, a coupling neither a control-only co-design (no sensing axis) nor a fixed-sensing pipeline (no shared-dynamics reshaping) can represent.

9.5 Computational complexity: ContEst with interior-point inner problems

Because the envelope theorem (Section 4) yields the design gradient from the *primal and dual of a single inner solve*—never by differentiating through the solver’s iterations—the per-design work is one inner solve plus a gradient contraction, and the outer L-BFGS iteration inherits the complexity

Table 6

MTDC sparse voltage-sensor allocation: the ℓ_1 sparsity/performance frontier (single-start sweep of λ_s , joint droop+sensing co-design). Across the co-designed rows J_{det} holds at its robust optimum, so the rising J_{est} isolates the price of removing sensors; the naive baseline telemeters all 25 at unit gain with over-provisioned droop $k = 3$.

λ_s	sensors	J_{est}	J_{det}	J_c
baseline	25	1.08	2.78	3.86
0.00	25	0.35	1.67	2.02
0.03	15	0.80	1.67	2.47
0.06	12	1.01	1.67	2.68
0.10	6	1.27	1.67	2.94
0.20	2	1.53	1.66	3.20

class of the inner solver. The gradient contraction is cheap: forming $\partial J / \partial \theta$ from the stored primal/dual costs $O(r_x^2 r_\theta)$ (the automatic-differentiation Jacobians of the model matrices times the dual blocks), dominated by the solve for every case we consider. The relevant question is therefore the cost of the inner solve.

Interior-point covariance/LMI solves scale as $O(r_x^6)$. Both ContEst inner problems place a covariance matrix among the decision variables. In the LQG SDPs of Section 5 the variable is the $r_x \times r_x$ covariance Σ (and its dual), i.e. $m = O(r_x^2)$ scalar unknowns; a primal–dual interior-point method assembles and factorizes a dense Schur-complement (Newton) system of size $m \times m$, at cost $O(m^3) = O(r_x^6)$ per iteration, times the $O(\sqrt{r_x}) - O(1)$ iterations interior-point methods take in practice. This is not the $O(r_x^3)$ of a single matrix factorization: the matrix-valued decision variable makes the *number* of unknowns quadratic in r_x , and it is that number, cubed, that governs the Newton solve. Empirically, solving the LQR covariance program with the Clarabel interior-point solver took ≈ 2.6 s at $r_x = 20$ and ≈ 30 s at $r_x = 30$ —an $(30/20)^6 \approx 11 \times$ growth, matching the $O(r_x^6)$ prediction—so a naive SDP inner problem caps ContEst at a few tens of states.

The eKF–MPC inner problem meets the same wall. There the decision trajectory is the information state $\zeta = [x; \text{vech } \Sigma]$ of dimension $r_x + \frac{r_x(r_x+1)}{2} = O(r_x^2)$; the horizon- N QP (23) thus has $O(Nr_x^2)$ variables and its interior-point solve again grows steeply in r_x (we measured ≈ 33 s per evaluation at $r_x = 32$ on the swing model). Consequently the nonlinear/constrained studies of Sections 9.1–9.3 are run at modest r_x , and large r_x on this path requires model reduction or structure exploitation (sparsity, chordal decomposition, first-order conic solvers).

The linear–quadratic case is $O(r_x^3)$ via Riccati. When the inner problem is linear and unconstrained—the LQG setting of Section 5, and, in its robust form, the MTDC study of Section 9.4—the *same* optimal value and envelope gradient are available without any interior-point step. The H_2 control cost $J_c = \text{tr}(PW)$ comes from the control Riccati equation;

the gradient needs only the stationary closed-loop covariance Σ_x from one discrete Lyapunov solve; and the estimation half comes from the dual (filter) Riccati. The robust (H_∞) output-feedback value used for MTDC — $J_{\text{det}} = \text{tr}(XW)$ and $J_{\text{est}} = \text{tr}(M\Sigma_e)$ — is identical in cost: both game Riccati (20) are DAREs in augmented data, so they use the *same* doubling and the *same* envelope identities (22). Each of these is $O(r_x^3)$, and structure-preserving doubling converges in a handful of iterations: the Riccati solve took ≈ 13 ms at $r_x = 200$ and ≈ 60 ms at $r_x = 400$ (the MTDC game Riccati at $r_x = 50$ solves in ≈ 2 ms), i.e. three-to-four orders of magnitude faster than the SDP at the crossover and scalable to the hundreds of states large power-system models require. For the H_2 case the interior-point covariance program and the Riccati recursion return the identical value and gradient (they agreed to 2×10^{-7} in our tests); we prefer the SDP form when its flexibility is needed—indefinite couplings, output-variance or chance constraints, or the joint estimation–control LMI—and the Riccati form, at $O(r_x^3)$, whenever the inner problem is a plain (or game-type) linear-quadratic regulator. The H_∞ case tips the balance further toward Riccati: at the minimal- γ optimum the bounded-real LMI (16) is rank-deficient and its dual ill-conditioned, whereas the fixed- γ game Riccati is well-posed—so for robust co-design we use the Riccati form throughout.

Takeaway. ContEst’s outer cost is (number of designs evaluated) $\times$ (one inner solve + $O(r_x^2 r_\theta)$). The envelope gradient keeps the per-design overhead negligible, so scalability is entirely a property of the inner solver: $O(r_x^6)$ for interior-point covariance/LMI or information-state MPC solves (practical to a few tens of states), versus $O(r_x^3)$ for the Riccati route in the linear-quadratic case (practical to hundreds). Choosing the inner solver to match the problem structure, rather than defaulting to a general-purpose conic solver, is what makes large-scale ContEst tractable.

9.6 Summary of improvements

Table 7 consolidates the four studies—their state/input/output dimensions, the design vector $\theta = (\theta_f, \theta_h)$ and where each axis enters ($\bar{f}$, $\bar{h}$, or the noise level V), the inner value used, and the estimation/control/total reductions—and Table 8 gives the absolute cost split for the three nonlinear (eKF–MPC) studies. In those three the total cost falls by 6–35%, with a reduction in the estimation cost J_{est} in every case, and—by construction—each optimum is an interior trade rather than a parameter pinned at a bound. The nature of the gain varies with the physics: for the spacecraft, a single *power/mass budget* is split across the star tracker, rate gyro and reaction wheel, favouring attitude sensing (J_{est} down 23.8%, the largest sensing gain of the closed-loop studies); for the column both axes are *placements*—feed streams and sensor trays—and the gain is estimation-led (J_{est} down 22.8% against a nearly flat J_{det}); and for the low-inertia grid the sensing axis is the *tuning of a dynamic estimator* (PLL bandwidth), which enters the measurement covariance $V(\theta)$ and, paired with the extra inverter damping it unlocks, drives

by far the largest total reduction (34.8%). The large-scale MTDC study is reported separately because its inner value is the robust H_∞ worst-case (guaranteed) cost, on a different numerical scale from the sampled closed-loop costs of the other three: there a single *control-side* droop axis reshapes the shared dynamics $A(\theta)$ and thereby improves *both* halves of the output-feedback design— J_{est} down 42.6%, J_{det} down 39.7%, J_{tot} down 43.4% (Table 7)—the design-level coupling a place-then-control pipeline cannot exploit. Every study compares against the no-upgrade / naive baseline θ_{nom} of its section (for the eKF–MPC studies, a certainty-equivalence controller on the eKF mean; Section 9), at a cost of a handful of inner convex solves per design.

10 Conclusion

We introduced ContEst, a co-design framework that brings state and parameter estimation into control co-design by optimizing actuation and sensing parameters against a single stochastic (dual control) objective expressed through the information state. The framework reduces, in the LQG regime, to a pair of semidefinite programs and, in the locally linearized regime, to a convex information-state MPC; sensor allocation enters as a mixed-integer selection with a sparse ℓ_1 relaxation. In every case the design gradient is obtained exactly and cheaply from the inner dual variables by the envelope theorem, sidestepping implicit differentiation of the solution map and differentiation of the KKT system. The prevalence of sequential design suggests broad applicability. Future work includes chance-constrained inner problems, information states beyond the eKF, and decomposition of the outer problem along MDO architectures for large interconnected systems.

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A Proof of the LQG regularity Proposition

This appendix proves Proposition 1 and derives the dual identities $S_{11}^* = P$, $S_{11}' = \Sigma^\varepsilon$, replacing the pointer to the sensitivity literature in Corollary 1 with a self-contained argument. We give the control SDP (9); the estimation SDP (10) is its formal dual (swap $A_\theta \leftrightarrow A_\theta^\top$, $B_\theta \leftrightarrow C_\theta^\top$, $W \leftrightarrow Q$, $R \leftrightarrow V$) and the same steps apply verbatim, yielding $P^{\varepsilon*} = \Sigma^\varepsilon$ and $S_{11}' = \Sigma^\varepsilon$.

(a) *Slater.* Pick any stabilizing K_0 (one exists by stabilizability) and let $\Sigma_0 \succ 0$ solve the strict Lyapunov inequality $A_{\text{cl},0}\Sigma_0 A_{\text{cl},0}^\top - \Sigma_0 + W \prec 0$ with $A_{\text{cl},0} = A_\theta + B_\theta K_0$; such Σ_0 exists because $A_{\text{cl},0}$ is Schur and $W \succ 0$. Setting $L_0 = K_0 \Sigma_0$ and $Z_0 \succ K_0 \Sigma_0 K_0^\top$ makes both LMIs of (9) strictly feasible, so Slater holds and strong duality with attainment follows [12].

Table 7

Consolidated view of the four studies: dimensions (r_x, r_u, r_y) ; the design vector $\theta = (\theta_f, \theta_h)$ as $n_f + n_h$ with where each axis enters ($\bar{f}$ dynamics/actuation, $\bar{h}$ output map, or the measurement covariance V); the inner value (eKF-MPC yields H_2 /sampled closed-loop costs, MTDC the H_∞ worst-case guaranteed costs); and the estimation/control/total reductions. MTDC's single control-side axis still improves J_{est} because θ_f reshapes the shared dynamics that the H_∞ filter must estimate through.

study	(r_x, r_u, r_y)	θ : $n_f + n_h$ (enters)	inner value	$J_{\text{est}} \downarrow$	$J_{\text{det}} \downarrow$	$J_{\text{tot}} \downarrow$
ADCS (spacecraft)	(6, 3, 6)	1+2: $e_{\text{rw}} \in \bar{f}$, $\alpha \in V$	eKF-MPC (H_2)	23.8%	4.0%	7.7%
Distillation	(12, 1, 3)	3+3: feeds $\in \bar{f}$, sensors $\in \bar{h}$	eKF-MPC (H_2)	22.8%	1.9%	6.5%
PLL/DSE (grid)	(9, 3, 6)	3+3: $e \in \bar{f}$, $b \in \bar{f}$, V	eKF-MPC (H_2)	20.9%	40.6%	34.8%
MTDC (HVDC)	(50, 25, 25)	25+0: $k \in \bar{f}$ (droop)	H_∞ output-fb.	42.6%	39.7%	43.4%

Table 8

Summary of cost reductions (baseline $\rightarrow$ ContEst optimum θ^*). J_{est} : estimation cost; J_{det} : control cost; $J_c = J_{\text{det}} + J_{\text{est}}$; $J_{\text{tot}} = J_c + J_{\text{des}}$. Percentages are reductions (larger is better). Baseline is the naive / no-upgrade θ_{nom} of each study (uniform budget for ADCS, naive default placement for distillation, nominal actuators and PLL for the low-inertia grid).

study	J_{est}		J_{det}		J_c		J_{des}	J_{tot}	
	base $\rightarrow$ opt	red.	base $\rightarrow$ opt	red.	base $\rightarrow$ opt	red.	opt	base $\rightarrow$ opt	red.
ADCS (spacecraft)	173.1 $\rightarrow$ 132.0	23.8%	448.3 $\rightarrow$ 430.5	4.0%	621.4 $\rightarrow$ 562.5	9.5%	11.0	621.4 $\rightarrow$ 573.5	7.7%
Distillation	12.8 $\rightarrow$ 9.9	22.8%	19.5 $\rightarrow$ 19.1	1.9%	32.3 $\rightarrow$ 29.0	10.2%	1.2	32.3 $\rightarrow$ 30.2	6.5%
PLL/DSE (grid)	26.6 $\rightarrow$ 21.0	20.9%	151.9 $\rightarrow$ 90.2	40.6%	178.5 $\rightarrow$ 111.2	37.7%	5.2	178.5 $\rightarrow$ 116.4	34.8%

(b) *Primal uniqueness.* With $W \succ 0$, the second LMI of (9) forces $\Sigma \succ 0$ and, taking Schur complements, encodes $A_{\text{cl}} \Sigma A_{\text{cl}}^\top - \Sigma + W \preceq 0$ for the gain $K = L \Sigma^{-1}$; a Lyapunov-eigenvector argument then shows any feasible (Σ, L) has Schur A_{cl} , i.e. encodes a stabilizing gain. The objective is the stationary cost $J(K) = \text{tr}(Q \Sigma_K) + \text{tr}(R K \Sigma_K K^\top)$, and completion of squares against the optimal K^* gives

$$J(K) - \text{tr}(PW) = \text{tr}((K - K^*)^\top (R + B_\theta^\top P B_\theta) (K - K^*) \Sigma_K), \quad (\text{A.1})$$

with P the stabilizing solution of the control DARE. Since $R + B_\theta^\top P B_\theta \succ 0$ and $\Sigma_K \succ 0$, the right-hand side is zero iff $K = K^*$, so the minimizer $\Sigma^* = \Sigma_{K^*}$ (and $L^* = K^* \Sigma^*$) is unique.

(c) *Dual uniqueness, strict complementarity, and $S_{11}^* = P$.* Let $S = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^\top & S_{22} \end{bmatrix} \succeq 0$ be the dual of the second LMI. Complementary slackness aligns the range of S with the kernel of the (rank-deficient at optimum) primal slack, $\ker \mathcal{G}^* = \text{range}[I; -A_{\text{cl}}^{*\top}]$, which parametrizes S through $S_{12} = -S_{11} A_{\text{cl}}^*$ (so $S_{12}^* = -P A_{\text{cl}}^*$) up to the free block S_{11} . Stationarity of the Lagrangian in Σ then reads

$$S_{11} = Q + K^{*\top} R K^* + A_{\text{cl}}^\top S_{11} A_{\text{cl}}, \quad (\text{A.2})$$

the closed-loop Stein (Lyapunov) equation, whose solution is unique because A_{cl} is Schur. But the control DARE is exactly $P = Q + K^{*\top} R K^* + A_{\text{cl}}^\top P A_{\text{cl}}$, so (A.2) gives $S_{11} = P$: the (1, 1) dual block is the control Riccati solution, and the dual is unique. Strict complementarity holds because $P \succeq Q \succ 0$ and $R \succ 0$ make the primal and dual slacks have complementary full rank. This verifies Assumption 1(A1)–(A3); (A4) is the modeling hypothesis that θ enters only through C^1 data, and Corollary 1 follows. $\square$

The identity $S_{11}^* = P$ (and dually $S_{11}'^* = \Sigma^\varepsilon$) is what makes the covariance-SDP gradient $\partial J_{\text{cont}} / \partial W = -S_{11}^*$ of (13) numerically identical to the Riccati gradient $\partial \text{tr}(PW) / \partial W = P$ used in Section 9.5.

B Proof of the fixed- γ H_∞ regularity Proposition

This appendix proves Proposition 2, the robust twin of Appendix A: at a fixed attenuation γ above the optimal one, the game GARE (20) plays exactly the role the DARE plays in the LQG case, so its stabilizing solution is a locally unique C^1 function of θ and $V_{\text{wc}}(\theta) = \text{tr}(X W_\theta)$ carries the closed-form gradient (22). The argument parallels Appendix A step for step, the only change being that the completion-of-squares now carries the *indefinite* game weight $\tilde{R} = \text{diag}(R, -\gamma^2 I)$; positive-definiteness of R is nowhere used, exactly as anticipated in Section 5.2. We treat the control GARE; the filter GARE follows by the transpose duality of Appendix A. Recall $\tilde{B}_\theta = [B_\theta \ E_\theta]$, $E_\theta = W_\theta^{1/2}$, the saddle gain $\tilde{K} = -(\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta$, and $A_{\text{cl}} = A_\theta + \tilde{B}_\theta \tilde{K}$.

(a) *Stabilizing solution.* For γ above the optimal attenuation γ^* , the discrete dynamic-game (equivalently, discrete bounded-real) Riccati theory gives a unique $X \succeq 0$ solving (20) for which $R + B_\theta^\top X B_\theta \succ 0$, the full game pivot $\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta$ is nonsingular with inertia (r_u, r_w) , and the game closed loop A_{cl} is Schur [7, 11]. This is the fixed- γ analog of the classical DARE existence invoked in Appendix A, and it is exactly the admissibility used in Section 5.2. (As γ^* is approached the pivot loses rank, which is the rank drop flagged in Remark 6; we stay strictly above it.)

(b) *Lyapunov (completion-of-squares) form.* Substituting $\tilde{K}$

into $Q + A_{\text{cl}}^\top X A_{\text{cl}} + \tilde{K}^\top \tilde{R} \tilde{K}$ and cancelling gives

$$X = Q + \tilde{K}^\top \tilde{R} \tilde{K} + A_{\text{cl}}^\top X A_{\text{cl}}, \quad (\text{B.1})$$

which is identical algebra to the LQG identity below (A.2), valid for the indefinite $\tilde{R}$ because completion of squares never uses its sign. This is the robust twin of $P = Q + K^\star^\top R K^\star + A_{\text{cl}}^\top P A_{\text{cl}}$.

(c) *Local uniqueness and C^1 dependence.* Write the GARE residual $\mathcal{R}(X, \theta) = Q + A_\theta^\top X A_\theta - A_\theta^\top X \tilde{B}_\theta (\tilde{R} + \tilde{B}_\theta^\top X \tilde{B}_\theta)^{-1} \tilde{B}_\theta^\top X A_\theta - X$. It is rational and C^1 in (X, θ) wherever the pivot is nonsingular, hence near a stabilizing root by (a). Because $\tilde{K}$ is the saddle point of the inner game (nonsingular pivot), the envelope of the game kills the multiplier's first-order variation and the partial Fréchet derivative in X collapses to the closed-loop Stein operator,

$$\partial_X \mathcal{R}(X, \theta)[\delta X] = A_{\text{cl}}^\top \delta X A_{\text{cl}} - \delta X, \quad (\text{B.2})$$

which is invertible precisely because A_{cl} is Schur (its eigenvalues are $\lambda_i \bar{\lambda}_j - 1 \neq 0$ for $|\lambda_i| < 1$). The implicit function theorem then yields a unique C^1 branch $\theta \mapsto X(\theta)$ through the stabilizing root, and with it C^1 maps $\tilde{K}(\theta)$, $A_{\text{cl}}(\theta)$, and the game closed-loop gramian $S(\theta)$ solving $S = A_{\text{cl}} S A_{\text{cl}}^\top + W_\theta$. This is the content of Proposition 2; it is the direct analog of Lemma-level smoothness of the DARE solution in Appendix A, and it needs neither an SDP dual nor strict complementarity.

(d) *C^1 value and the closed-form gradient.* Since $V_{\text{wc}} = \text{tr}(X W_\theta)$ with $X \in C^1$ and $W_\theta \in C^1$, $V_{\text{wc}} \in C^1$. Differentiating (B.1), the saddle stationarity of $\tilde{K}$ annihilates all $d\tilde{K}$ terms, so dX solves the Stein equation $dX - A_{\text{cl}}^\top dX A_{\text{cl}} = \Xi$ with source $\Xi = dQ + \tilde{K}^\top d\tilde{R} \tilde{K} + A_{\text{cl}}^\top X (dA_\theta + d\tilde{B}_\theta \tilde{K}) + (\cdot)^\top$. Contracting with the gramian S and using $\text{tr}(W_\theta dX) = \text{tr}(S(dX - A_{\text{cl}}^\top dX A_{\text{cl}})) = \text{tr}(S\Xi)$ gives, term by term,

$$\begin{aligned} \frac{\partial V_{\text{wc}}}{\partial A_\theta} &= 2 X A_{\text{cl}} S, & \frac{\partial V_{\text{wc}}}{\partial \tilde{B}_\theta} &= 2 X A_{\text{cl}} S \tilde{K}^\top, \\ \frac{\partial V_{\text{wc}}}{\partial Q} &= S, & \frac{\partial V_{\text{wc}}}{\partial \tilde{R}} &= \tilde{K} S \tilde{K}^\top, \end{aligned} \quad (\text{B.3})$$

together with the explicit $\partial V_{\text{wc}}/\partial W_\theta = X$ from the objective; these are exactly (22). The disturbance channel $E_\theta = W_\theta^{1/2}$ enters through the E -block of $\partial V_{\text{wc}}/\partial \tilde{B}_\theta$, so when a single design coordinate θ_i moves both the objective weight W_θ and the channel E_θ its gradient sums the two contributions, precisely the data-object chain rule (12) of Assumption 1(A4).

(e) *Filter GARE by duality.* Under the transpose substitution of Appendix A— $A_\theta \mapsto A_\theta^\top$, $\tilde{B}_\theta \mapsto [C_\theta^\top \ L_e^\top]$, $\tilde{R} \mapsto \text{diag}(V, -\gamma^2 I)$, W_θ the driving covariance—the worst-case a-priori error covariance Σ_e solves the control GARE on the

dual data, so (a)–(d) transfer verbatim: $\Sigma_e(\theta)$ is a locally unique C^1 stabilizing solution and $V_{\text{est}}(\theta) = \text{tr}(M \Sigma_e)$ has the dual gradient (contract $2 \Sigma_e \tilde{A}_{\text{cl}} \tilde{S}$ against $\partial A_\theta^\top / \partial \theta$, with the C_θ, V blocks of $2 \Sigma_e \tilde{A}_{\text{cl}} \tilde{S} \tilde{K}^\top$), matching Section 5.2.

(f) *The H_2 limit.* As $\gamma \rightarrow \infty$ the disturbance player is switched off: the $-\gamma^2 I$ pivot block deactivates, the GARE (20) reduces to the control DARE $P = Q + K^\star^\top R K^\star + A_{\text{cl}}^\top P A_{\text{cl}}$ of Appendix A, $X \rightarrow P$, $S \rightarrow \Sigma^\star$, and (B.3) collapses to the LQG gradients of Appendix A. The H_2 /LQG result is thus the $\gamma \rightarrow \infty$ boundary of this one, and (unlike the minimal- γ BRL SDP of Remark 6) no rank drop or ill-conditioned dual ever enters the fixed- γ route. $\square$

References

- [1] Akshay Agrawal, Brandon Amos, Shane Barratt, Stephen Boyd, Steven Diamond, and J. Zico Kolter. Differentiable convex optimization layers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [2] Farid Alizadeh, Jean-Pierre A. Haeberly, and Michael L. Overton. Complementarity and nondegeneracy in semidefinite programming. *Mathematical Programming*, 77(1):111–128, 1997.
- [3] James T. Allison and Daniel R. Herber. Multidisciplinary design optimization of dynamic engineering systems. *AIAA Journal*, 52(4):691–710, 2014.
- [4] Brandon Amos and J. Zico Kolter. OptNet: Differentiable optimization as a layer in neural networks. In *International Conference on Machine Learning (ICML)*, 2017.
- [5] Brian DO Anderson and John B Moore. *Optimal filtering*. Courier Corporation, 2012.
- [6] Yaakov Bar-Shalom and Edison Tse. Dual effect, certainty equivalence, and separation in stochastic control. *IEEE Transactions on Automatic Control*, 19(5):494–500, 1974.
- [7] Tamer Başar and Pierre Bernhard. *H_∞ -Optimal Control and Related Minimax Design Problems: A Dynamic Game Approach*. Birkhäuser, Boston, MA, 2 edition, 1995.
- [8] Amir Beck. *First-Order Methods in Optimization*. MOS-SIAM Series on Optimization. Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2017.
- [9] Lorenz T. Biegler. *Nonlinear Programming: Concepts, Algorithms, and Applications to Chemical Processes*. SIAM, 2010.
- [10] J. Frédéric Bonnans and Alexander Shapiro. *Perturbation Analysis of Optimization Problems*. Springer, 2000.
- [11] Stephen Boyd, Laurent El Ghaoui, Eric Feron, and Venkataramanan Balakrishnan. *Linear matrix inequalities in system and control theory*. SIAM, 1994.
- [12] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004.
- [13] Emmanuel J. Candès, Michael B. Wakin, and Stephen P. Boyd. Enhancing sparsity by reweighted ℓ_1 minimization. *Journal of Fourier Analysis and Applications*, 14(5):877–905, 2008.
- [14] Niladri Das and Raktim Bhattacharya. Sparse sensing architecture for Kalman filtering with guaranteed error bound. In *1st IAA Conference on Space Situational Awareness (ICSSA)*, 2017.
- [15] Neil K. Dhingra, Mihailo R. Jovanović, and Zhi-Quan Luo. An ADMM algorithm for optimal sensor and actuator selection. In *53rd IEEE Conference on Decision and Control (CDC)*, pages 4039–4044, 2014.

- [16] Nikolaos A. Dangelakis, Baris Burnak, Justin Katz, and Efstratios N. Pistikopoulos. Process design and control optimization: A simultaneous approach by multi-parametric programming. *AIChE Journal*, 63(11):4827–4846, 2017.
- [17] Hosam K. Fathy, Julie A. Reyer, Panos Y. Papalambros, and A. Galip Ulsoy. On the coupling between the plant and controller optimization problems. In *Proceedings of the American Control Conference (ACC)*, volume 3, pages 1864–1869, 2001.
- [18] Aleksandr Aronovich Feldbaum. Dual control theory. i. *Avtomatika i Telemekhanika*, 21(9):1240–1249, 1960.
- [19] Mario Garcia-Sanz. Control co-design: an engineering game changer. *Advanced Control for Applications: Engineering and Industrial Systems*, 1(1):e18, 2019.
- [20] Michel Gevers. Identification for control: From the early achievements to the revival of experiment design. *European journal of control*, 11(4-5):335–352, 2005.
- [21] Stephen Gould, Basura Fernando, Anoop Cherian, Peter Anderson, Rodrigo Santa Cruz, and Edison Guo. On differentiating parameterized argmin and argmax problems with application to bilevel optimization, 2016. arXiv:1607.05447.
- [22] Karolos M. Grigoriadis, Guoming Zhu, and Robert E. Skelton. Optimal redesign of linear systems. *Journal of Dynamic Systems, Measurement, and Control*, 118(3):598–605, 1996.
- [23] Arthur L. Hale, Ronald J. Lisowski, and Wayne E. Dahl. Optimal simultaneous structural and control design of maneuvering flexible spacecraft. *Journal of Guidance, Control, and Dynamics*, 8(1):86–93, 1985.
- [24] Daniel R. Herber and James T. Allison. Nested and simultaneous solution strategies for general combined plant and control design problems. *Journal of Mechanical Design*, 141(1):011402, 2019.
- [25] Siddharth Joshi and Stephen Boyd. Sensor selection via convex optimization. *IEEE Transactions on Signal Processing*, 57(2):451–462, 2009.
- [26] P. R. Kumar and Pravin Varaiya. *Stochastic Systems: Estimation, Identification, and Adaptive Control*. Classics in Applied Mathematics. SIAM, 2015.
- [27] Faming Li, Mauricio C. de Oliveira, and Robert E. Skelton. Integrating information architecture and control or estimation design. *SICE Journal of Control, Measurement, and System Integration*, 1(2):120–128, 2008.
- [28] Ali Mesbah. Stochastic model predictive control with active uncertainty learning: A survey on dual control. *Annual Reviews in Control*, 45:107–117, 2018.
- [29] Paul Milgrom and Ilya Segal. Envelope theorems for arbitrary choice sets. *Econometrica*, 70(2):583–601, 2002.
- [30] Junjiro Onoda and Raphael T. Haftka. An approach to structure/control simultaneous optimization for large flexible spacecraft. *AIAA Journal*, 25(8):1133–1138, 1987.
- [31] Mohammad S Ramadan and Mihai Anitescu. Extended Kalman filter – Koopman operator for tractable stochastic optimal control. *IEEE Control Systems Letters*, 2024.
- [32] Sidney Resnick. *A probability path*. Springer, 2019.
- [33] Vassilis Sakizlis, John D. Perkins, and Efstratios N. Pistikopoulos. Recent advances in optimization-based simultaneous process and control design. *Computers & Chemical Engineering*, 28(10):2069–2086, 2004.
- [34] Rajeev Saraf, Raktim Bhattacharya, and Robert E. Skelton. H2 optimal sensing architecture with model uncertainty. In *American Control Conference (ACC)*, 2017.
- [35] Alexander Shapiro. First and second order analysis of nonlinear semidefinite programs. *Mathematical Programming*, 77(1):301–320, 1997.
- [36] Mahdi Sharifzadeh. Integration of process design and control: A review. *Chemical Engineering Research and Design*, 91(12):2515–2549, 2013.
- [37] Milad Siami and Ali Jadbabaie. A separation theorem for joint sensor and actuator scheduling with guaranteed performance bounds. *Automatica*, 119:109054, 2020.
- [38] Sigurd Skogestad and Ian Postlethwaite. *Multivariable Feedback Control: Analysis and Design*. Wiley, 2 edition, 2005.
- [39] Vasileios Tzoumas, Luca Carlone, George J. Pappas, and Ali Jadbabaie. LQG control and sensing co-design. *IEEE Transactions on Automatic Control*, 66(4):1468–1483, 2021.
- [40] Vasileios Tzoumas, Ali Jadbabaie, and George J. Pappas. Sensor placement for optimal Kalman filtering: Fundamental limits, submodularity, and algorithms. In *American Control Conference (ACC)*, pages 191–196, 2016.
- [41] Gioele Zardini, Andrea Censi, and Emilio Frazzoli. Co-design of autonomous systems: From hardware selection to control synthesis. In *European Control Conference (ECC)*, pages 682–689, 2021.
- [42] Haotian Zhang, Raid Ayoub, and Shreyas Sundaram. Sensor selection for Kalman filtering of linear dynamical systems: Complexity, limitations and greedy algorithms. *Automatica*, 78:202–210, 2017.
- [43] Junbo Zhao, Antonio Gomez-Exposito, Marcos Netto, Lamine Mili, Ali Abur, Vladimir Terzija, Innocent Kamwa, Bikash Pal, Abhinav Kumar Singh, Junjian Qi, Zhenyu Huang, and A. P. Sakis Meliopoulos. Power system dynamic state estimation: Motivations, definitions, methodologies, and future work. *IEEE Transactions on Power Systems*, 34(4):3188–3198, 2019.